

STAT394_Group6_Codes

2026-09-18

```
##          Temp      pH      EC      TS      CA2+      Mg2+      NA+      k+      Cl-      SO2-4
## Temp      1.000 -0.048  0.014 -0.019 -0.363 -0.349  0.139 -0.055  0.130  0.319
## pH       -0.048  1.000 -0.301 -0.277 -0.447 -0.476 -0.197 -0.460 -0.112 -0.070
## EC        0.014 -0.301  1.000  0.960  0.129  0.153  0.592  0.444  0.277  0.239
## TS       -0.019 -0.277  0.960  1.000  0.007  0.127  0.488  0.375  0.250 -0.006
## CA2+     -0.363 -0.447  0.129  0.007  1.000  0.416  0.110  0.498 -0.002  0.295
## Mg2+     -0.349 -0.476  0.153  0.127  0.416  1.000 -0.100  0.188 -0.097 -0.054
## NA+       0.139 -0.197  0.592  0.488  0.110 -0.100  1.000  0.651  0.665  0.209
## k+       -0.055 -0.460  0.444  0.375  0.498  0.188  0.651  1.000  0.142  0.155
## Cl-       0.130 -0.112  0.277  0.250 -0.002 -0.097  0.665  0.142  1.000 -0.037
## SO2-4     0.319 -0.070  0.239 -0.006  0.295 -0.054  0.209  0.155 -0.037  1.000
## HCO-3    -0.045 -0.360  0.623  0.530  0.320  0.230  0.877  0.827  0.350  0.130
## SiO2      0.370 -0.006  0.156  0.154 -0.093 -0.069  0.214  0.372 -0.159  0.135
##          HCO-3      SiO2
## Temp     -0.045  0.370
## pH       -0.360 -0.006
## EC        0.623  0.156
## TS        0.530  0.154
## CA2+      0.320 -0.093
## Mg2+      0.230 -0.069
## NA+       0.877  0.214
## k+        0.827  0.372
## Cl-       0.350 -0.159
## SO2-4     0.130  0.135
## HCO-3     1.000  0.314
## SiO2      0.314  1.000

## Standard deviations (1, ..., p=12):
## [1] 2.05162266 1.45109353 1.21822561 1.08087635 1.00623083 0.93965377
## [7] 0.70462921 0.57618689 0.44497244 0.32543353 0.05580552 0.04289133
##
## Rotation (n x k) = (12 x 12):
##          PC1          PC2          PC3          PC4          PC5          PC6
## Temp -0.008933291 -0.47274040 -0.359100401 -0.05024999 -0.25536433  0.47387530
## pH    0.255691665 -0.30221908  0.138241266  0.01110620  0.22149343 -0.60932785
## EC    -0.393539305 -0.09216584  0.167859022  0.30807498 -0.36905982 -0.20134825
## TS    -0.350425860 -0.09866922  0.276988222  0.44622358 -0.28642870 -0.09759908
## CA2+  -0.194535218  0.46993907 -0.228318890 -0.31475872 -0.02990099 -0.20702531
## Mg2+  -0.126284300  0.50743717 -0.014287074  0.19261770 -0.10317612  0.27982733
## NA+   -0.408188550 -0.24495977  0.118678499 -0.27671475  0.16959131 -0.04762706
## k+    -0.398677248  0.09807998 -0.212841166 -0.04849194  0.36066956 -0.04696439
## Cl-   -0.209237698 -0.21442117  0.406428652 -0.50455285 -0.02988726  0.27462350
## SO2-4 -0.118436488 -0.09385594 -0.495770458 -0.29202119 -0.52379559 -0.38067730
## HCO-3 -0.447051993 -0.01156696 -0.003401385 -0.04231818  0.28030934 -0.05740156
## SiO2  -0.140905230 -0.25312177 -0.473638447  0.38286811  0.37851241  0.06949860
```

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##          PC7          PC8          PC9          PC10          PC11
## Temp -0.03025873  0.06495604 -0.58594277 -0.07642163  0.029759888
## pH    0.39728101 -0.06942416 -0.47764880  0.11291601  0.006807658
## EC    -0.02825150 -0.07672586 -0.04082105 -0.02708020  0.274621336
## TS    -0.14652416 -0.17414789 -0.11074046  0.07774116 -0.235917138
## CA2+  -0.18007566 -0.52331078 -0.36820576 -0.30633364  0.082378988
## Mg2+   0.71539242  0.11761350 -0.17903927  0.13761066  0.132746236
## NA+    0.10770341  0.21026834  0.12375787 -0.23562865  0.643661458
## k+    -0.27181928  0.16502403 -0.19326676  0.71382042  0.022629254
## Cl-    0.25274942 -0.45265201  0.09649352  0.25837171 -0.236606996
## SO2-4  0.22768241  0.13071816  0.29897702  0.16970005 -0.167273908
## HCO-3  0.13516194  0.35291574 -0.08651678 -0.44425734 -0.586776514
## SiO2   0.23514918 -0.49908240  0.29683477 -0.06226230  0.022452293
##          PC12
## Temp   0.0002824031
## pH     0.0107231096
## EC     -0.6725307280
## TS     0.6118008796
## CA2+   0.0622391153
## Mg2+   0.0779287025
## NA+    0.3327790340
## k+    -0.0545287327
## Cl-    -0.1155338902
## SO2-4  0.1083035842
## HCO-3 -0.1560941039
## SiO2  -0.0134931512

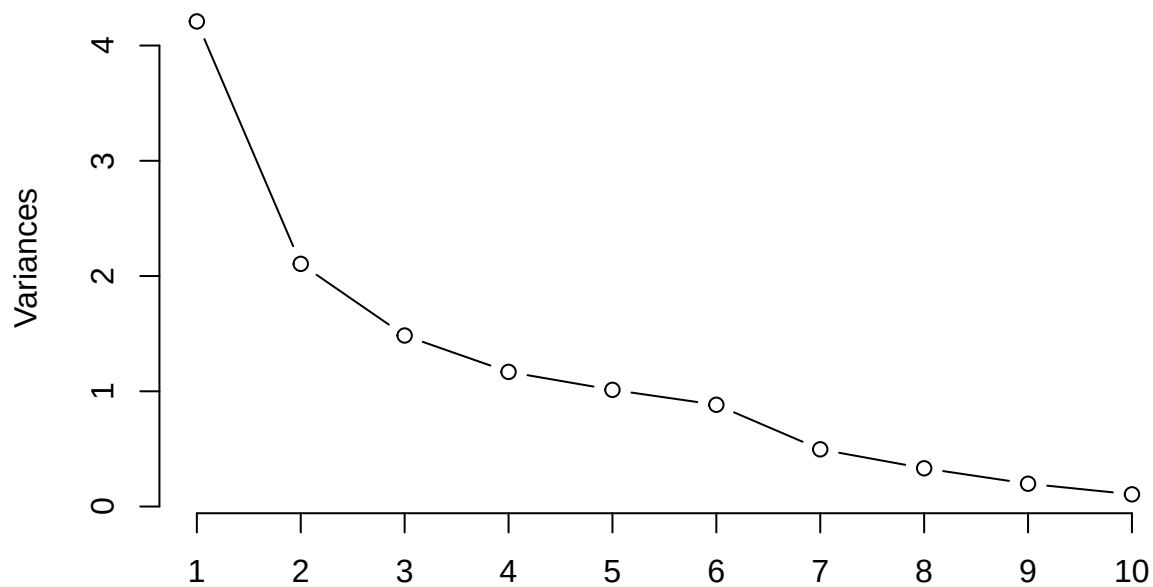
## Importance of components:
##          PC1    PC2    PC3    PC4    PC5    PC6    PC7
## Standard deviation  2.0516 1.4511 1.2182 1.08088 1.00623 0.93965 0.70463
## Proportion of Variance 0.3508 0.1755 0.1237 0.09736 0.08438 0.07358 0.04138
## Cumulative Proportion 0.3508 0.5262 0.6499 0.74727 0.83164 0.90522 0.94660
##          PC8    PC9    PC10    PC11    PC12
## Standard deviation  0.57619 0.4450 0.32543 0.05581 0.04289
## Proportion of Variance 0.02767 0.0165 0.00883 0.00026 0.00015
## Cumulative Proportion 0.97426 0.9908 0.99959 0.99985 1.00000

##          PC1    PC2    PC3    PC4    PC5    PC6    PC7    PC8    PC9    PC10
## Temp -0.009 -0.473 -0.359 -0.050 -0.255  0.474 -0.030  0.065 -0.586 -0.076
## pH    0.256 -0.302  0.138  0.011  0.221 -0.609  0.397 -0.069 -0.478  0.113
## EC    -0.394 -0.092  0.168  0.308 -0.369 -0.201 -0.028 -0.077 -0.041 -0.027
## TS    -0.350 -0.099  0.277  0.446 -0.286 -0.098 -0.147 -0.174 -0.111  0.078
## CA2+  -0.195  0.470 -0.228 -0.315 -0.030 -0.207 -0.180 -0.523 -0.368 -0.306
## Mg2+  -0.126  0.507 -0.014  0.193 -0.103  0.280  0.715  0.118 -0.179  0.138
## NA+   -0.408 -0.245  0.119 -0.277  0.170 -0.048  0.108  0.210  0.124 -0.236
## k+    -0.399  0.098 -0.213 -0.048  0.361 -0.047 -0.272  0.165 -0.193  0.714
## Cl-   -0.209 -0.214  0.406 -0.505 -0.030  0.275  0.253 -0.453  0.096  0.258
## SO2-4 -0.118 -0.094 -0.496 -0.292 -0.524 -0.381  0.228  0.131  0.299  0.170
## HCO-3 -0.447 -0.012 -0.003 -0.042  0.280 -0.057  0.135  0.353 -0.087 -0.444
## SiO2  -0.141 -0.253 -0.474  0.383  0.379  0.069  0.235 -0.499  0.297 -0.062
##          PC11    PC12
## Temp  0.030  0.000
## pH    0.007  0.011
## EC    0.275 -0.673

```

```
## TS      -0.236  0.612
## CA2+    0.082  0.062
## Mg2+    0.133  0.078
## NA+     0.644  0.333
## k+      0.023 -0.055
## Cl-     -0.237 -0.116
## SO2-4   -0.167  0.108
## HCO-3   -0.587 -0.156
## SiO2     0.022 -0.013
```

PCA.X



```
##
## Call:
## factanal(x = X, factors = 4)
##
## Uniquenesses:
## Temp    pH    EC    TS    CA2+  Mg2+  NA+    k+    Cl-  SO2-4  HCO-3  SiO2
## 0.789 0.783 0.005 0.005 0.632 0.560 0.005 0.274 0.280 0.005 0.005 0.876
##
## Loadings:
##      Factor1 Factor2 Factor3 Factor4
## Temp  -0.107         0.252  0.369
## pH    -0.376 -0.220  0.162
## EC     0.330  0.924  0.114  0.140
## TS     0.199  0.968  0.103
## CA2+   0.489         -0.315  0.170
## Mg2+   0.369  0.109 -0.509 -0.183
```

```

## NA+      0.652   0.307   0.681   0.108
## k+       0.825   0.194
## Cl-      0.164   0.830
## SO2-4    0.221           0.969
## HCO-3    0.893   0.328   0.294
## SiO2     0.337
##
##              Factor1 Factor2 Factor3 Factor4
## SS loadings      2.747   2.126   1.726   1.184
## Proportion Var    0.229   0.177   0.144   0.099
## Cumulative Var    0.229   0.406   0.550   0.649
##
## Test of the hypothesis that 4 factors are sufficient.
## The chi square statistic is 94.44 on 24 degrees of freedom.
## The p-value is 2.63e-10

```

```

##      No.      Name      T      pH
## Min.   : 1.00   Length:30   Min.   :14.40   Min.   :6.500
## 1st Qu.: 8.25   Class :character 1st Qu.:25.12   1st Qu.:6.825
## Median :15.50   Mode  :character  Median :30.50   Median :7.250
## Mean   :15.50           Mean   :34.07   Mean   :7.303
## 3rd Qu.:22.75           3rd Qu.:39.77   3rd Qu.:7.500
## Max.   :30.00           Max.   :95.00   Max.   :9.000
##      EC      TS      Ca2.      Mg2.
## Min.   : 110.0   Min.   :0.1000   Min.   : 4.00   Min.   : 0.10
## 1st Qu.: 419.8   1st Qu.:0.3977   1st Qu.: 32.00   1st Qu.: 18.07
## Median :1100.0   Median :1.0450   Median : 50.00   Median : 27.75
## Mean   :1433.7   Mean   :1.4420   Mean   : 61.02   Mean   : 44.46
## 3rd Qu.:2174.2   3rd Qu.:2.0642   3rd Qu.: 78.50   3rd Qu.: 60.25
## Max.   :5100.0   Max.   :6.2000   Max.   :178.00   Max.   :293.00
##      Na.      K.      Cl.      SO2.4
## Min.   : 6.40   Min.   : 0.20   Min.   : 7.00   Min.   : 0.90
## 1st Qu.: 53.55   1st Qu.: 3.25   1st Qu.: 18.00   1st Qu.: 5.25
## Median : 253.20   Median : 9.15   Median : 33.25   Median : 17.00
## Mean   : 447.09   Mean   :16.58   Mean   :108.70   Mean   : 57.53
## 3rd Qu.: 468.75   3rd Qu.:30.75   3rd Qu.: 73.00   3rd Qu.: 54.50
## Max.   :2280.00   Max.   :47.00   Max.   :1560.00   Max.   :486.00
##      HCO.3      SiO2      Geological.structure      X
## Min.   : 79.3   Min.   : 2.00   Length:30   Length:30
## 1st Qu.: 384.5   1st Qu.: 20.00   Class :character   Class :character
## Median : 859.5   Median : 35.00   Mode  :character   Mode  :character
## Mean   :1296.9   Mean   : 38.73
## 3rd Qu.:2111.2   3rd Qu.: 45.00
## Max.   :3538.0   Max.   :120.00

```

```

##      T pH EC TS Ca2. Mg2. Na. K. Cl. SO2.4 HCO.3 SiO2
## 1 45.0 7.7 3770 4.173 5.6 3.8 1157.0 7.9 56.8 2 2903.6 36.7
## 2 25.0 7.3 5100 6.200 24.0 24.0 476.0 4.0 369.0 2 829.0 25.0
## 3 25.0 6.6 677 0.641 120.0 36.6 200.0 22.2 89.0 55 890.0 18.0
## 4 48.0 7.0 2219 2.119 60.0 18.0 2280.0 30.0 1560.0 2 3360.0 20.0
## 5 14.4 7.5 1100 1.100 56.0 48.8 226.1 3.5 198.8 82 573.4 22.0
## 6 33.0 7.4 722 0.538 60.0 67.0 69.0 7.1 28.0 90 555.0 10.0

```

```

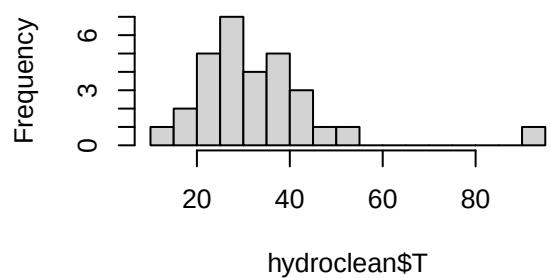
##      T      pH      EC      TS

```

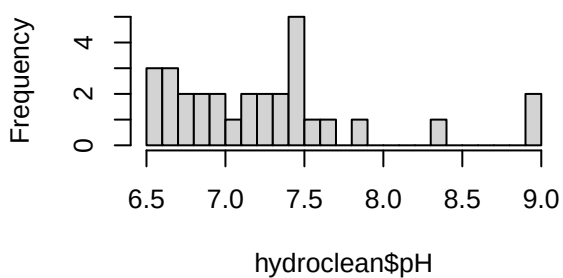
##	Min.	:14.40	Min.	:6.500	Min.	: 110.0	Min.	:0.1000
##	1st Qu.:	25.12	1st Qu.:	6.825	1st Qu.:	419.8	1st Qu.:	0.3977
##	Median	:30.50	Median	:7.250	Median	:1100.0	Median	:1.0450
##	Mean	:34.07	Mean	:7.303	Mean	:1433.7	Mean	:1.4420
##	3rd Qu.:	39.77	3rd Qu.:	7.500	3rd Qu.:	2174.2	3rd Qu.:	2.0642
##	Max.	:95.00	Max.	:9.000	Max.	:5100.0	Max.	:6.2000
##	Ca2.		Mg2.		Na.		K.	
##	Min.	: 4.00	Min.	: 0.10	Min.	: 6.40	Min.	: 0.20
##	1st Qu.:	32.00	1st Qu.:	18.07	1st Qu.:	53.55	1st Qu.:	3.25
##	Median	: 50.00	Median	: 27.75	Median	: 253.20	Median	: 9.15
##	Mean	: 61.02	Mean	: 44.46	Mean	: 447.09	Mean	:16.58
##	3rd Qu.:	78.50	3rd Qu.:	60.25	3rd Qu.:	468.75	3rd Qu.:	30.75
##	Max.	:178.00	Max.	:293.00	Max.	:2280.00	Max.	:47.00
##	Cl.		SO2.4		HCO.3		SiO2	
##	Min.	: 7.00	Min.	: 0.90	Min.	: 79.3	Min.	: 2.00
##	1st Qu.:	18.00	1st Qu.:	5.25	1st Qu.:	384.5	1st Qu.:	20.00
##	Median	: 33.25	Median	: 17.00	Median	: 859.5	Median	: 35.00
##	Mean	: 108.70	Mean	: 57.53	Mean	:1296.9	Mean	: 38.73
##	3rd Qu.:	73.00	3rd Qu.:	54.50	3rd Qu.:	2111.2	3rd Qu.:	45.00
##	Max.	:1560.00	Max.	:486.00	Max.	:3538.0	Max.	:120.00

The multivariate normal distribution is a well-established and mathematically convenient model that underlies many statistical procedures. So it is a reasonable and important step to understand our data, a step which might decide our subsequent analysis. Tanaskovic et al. tested whether the individual variables followed a normal distribution before their analysis, and ultimately transformed the variables which failed the Shapiro-Wilk tests for normality (H_0 : the data comes from a normally distributed population) using a Box-Cox transformation. It is simple to attempt this ourselves; we can even look at histograms and QQ-plots to visually see the same.

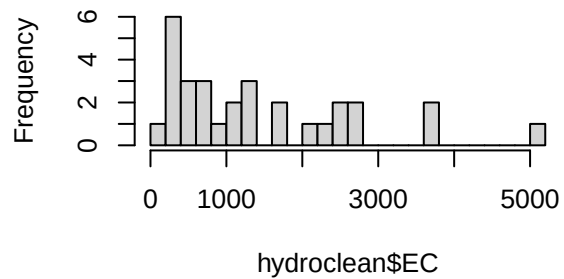
Histogram of hydroclean\$T



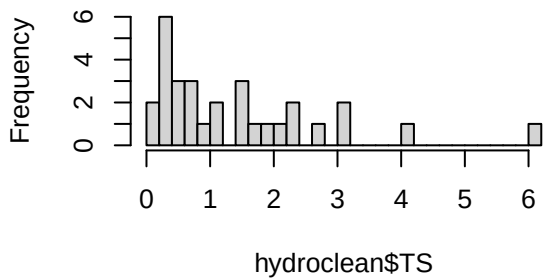
Histogram of hydroclean\$pH



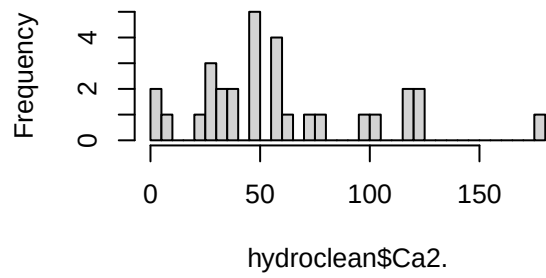
Histogram of hydroclean\$EC



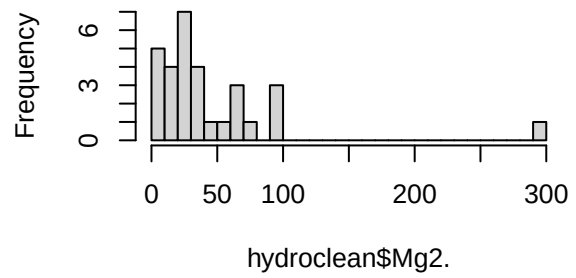
Histogram of hydroclean\$TS



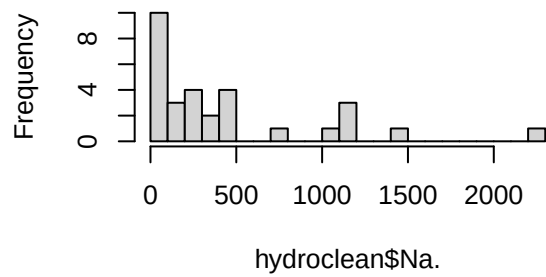
Histogram of hydroclean\$Ca2.



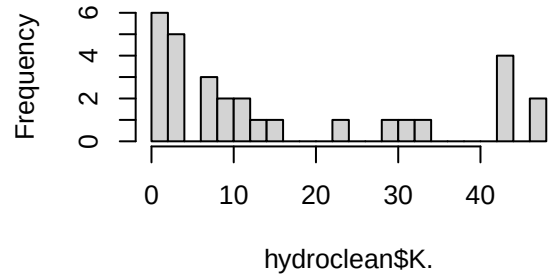
Histogram of hydroclean\$Mg2.



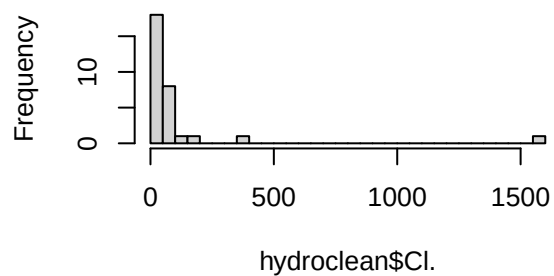
Histogram of hydroclean\$Na.



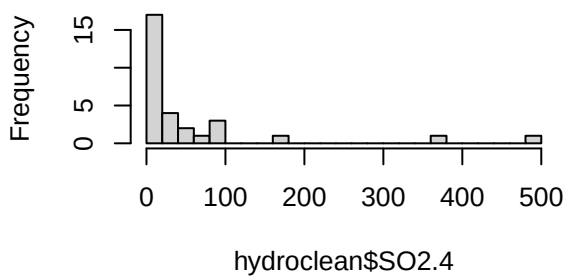
Histogram of hydroclean\$K.



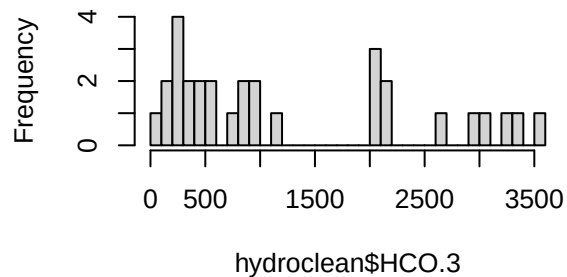
Histogram of hydroclean\$Cl.



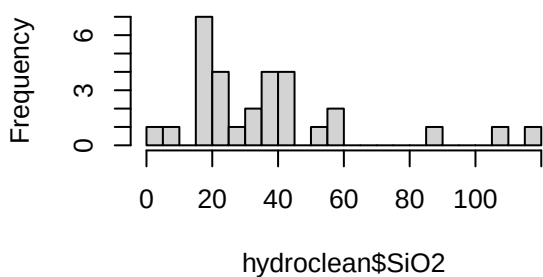
Histogram of hydroclean\$SO2.4

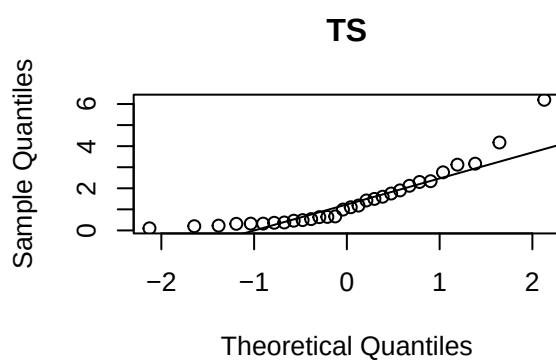
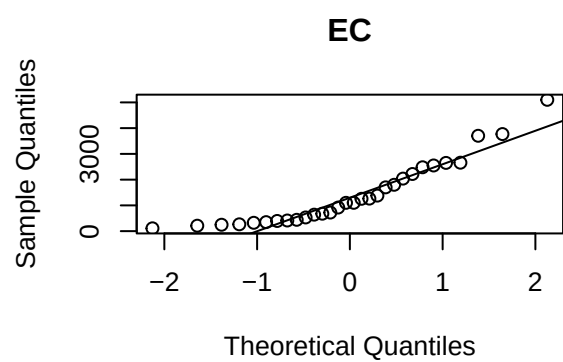
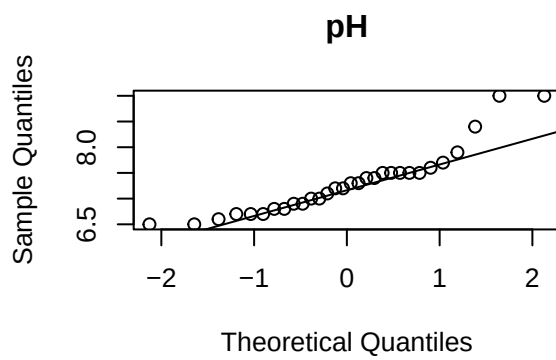
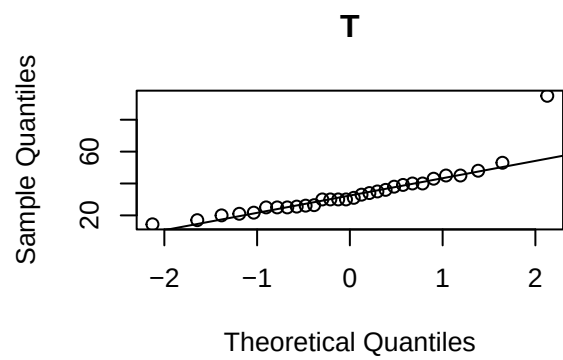


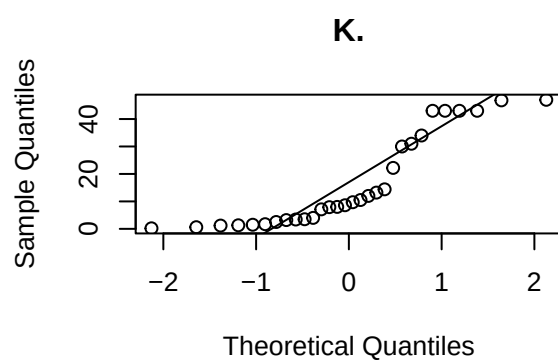
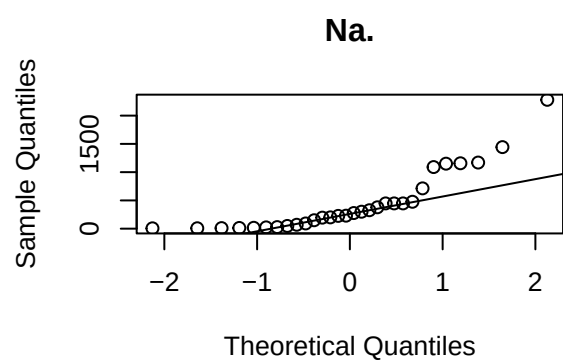
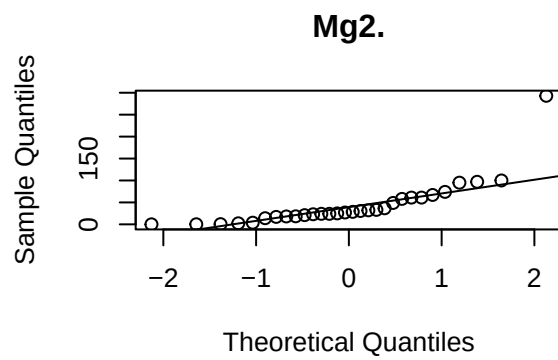
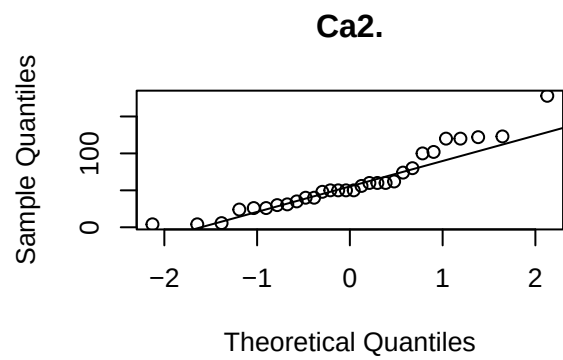
Histogram of hydroclean\$HCO.3

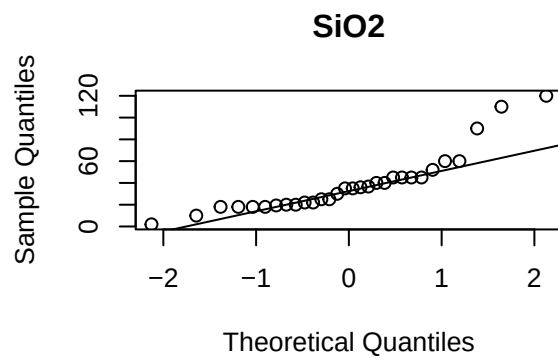
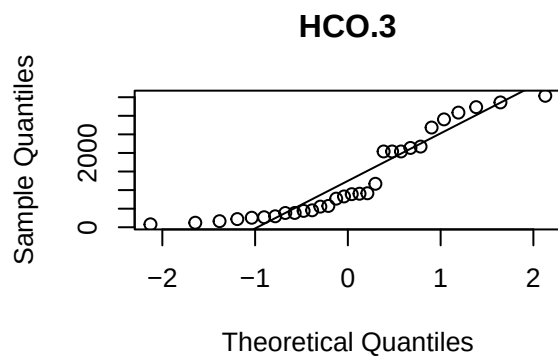
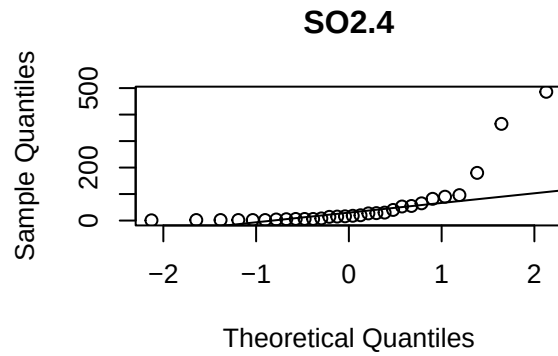
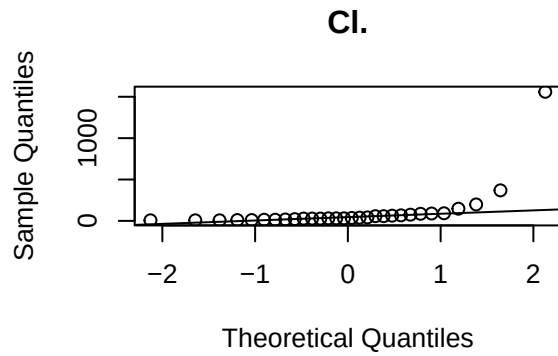


Histogram of hydroclean\$SiO2









```
##
##  Shapiro-Wilk normality test
##
## data:  hydroclean$T
## W = 0.79825, p-value = 6.092e-05

##
##  Shapiro-Wilk normality test
##
## data:  hydroclean$pH
## W = 0.87966, p-value = 0.002769

##
##  Shapiro-Wilk normality test
##
## data:  hydroclean$EC
## W = 0.87059, p-value = 0.001726

##
##  Shapiro-Wilk normality test
##
## data:  hydroclean$TS
## W = 0.82454, p-value = 0.00019

##
```

```

## Shapiro-Wilk normality test
##
## data:  hydroclean$Ca2.
## W = 0.91774, p-value = 0.02344

##
## Shapiro-Wilk normality test
##
## data:  hydroclean$Mg2.
## W = 0.64505, p-value = 2.714e-07

##
## Shapiro-Wilk normality test
##
## data:  hydroclean$Na.
## W = 0.77634, p-value = 2.501e-05

##
## Shapiro-Wilk normality test
##
## data:  hydroclean$K.
## W = 0.81198, p-value = 0.0001092

##
## Shapiro-Wilk normality test
##
## data:  hydroclean$Cl.
## W = 0.34031, p-value = 1.566e-10

##
## Shapiro-Wilk normality test
##
## data:  hydroclean$SO2.4
## W = 0.54516, p-value = 1.664e-08

##
## Shapiro-Wilk normality test
##
## data:  hydroclean$HCO.3
## W = 0.85388, p-value = 0.0007483

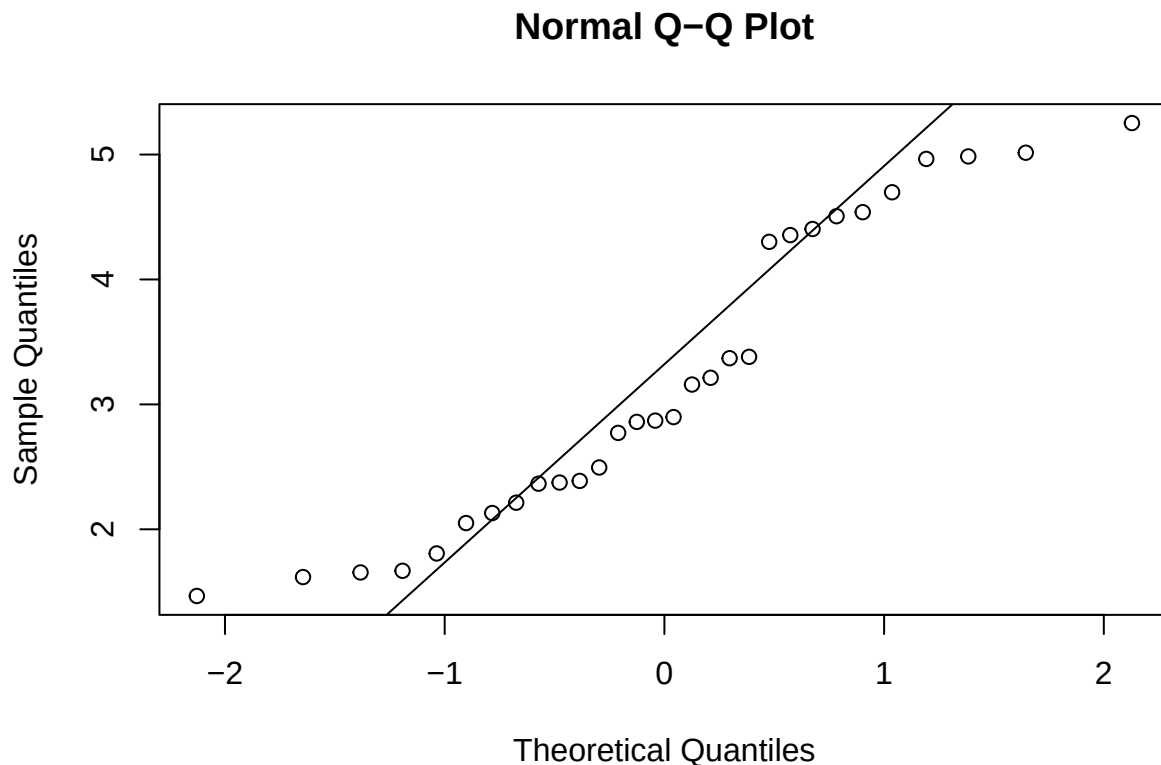
##
## Shapiro-Wilk normality test
##
## data:  hydroclean$SiO2
## W = 0.83043, p-value = 0.0002479

```

The histograms and QQ-plots show the individual variables deviate from normality to varying degrees. Understandably, for what the variables are, we see the measures are bounded below by zero and generally right skewed, where most observations are relatively low with a few very high values. Obviously these characteristics are different from the unbounded, symmetric normal distribution.

Just like Tanaskovic, our Shapiro-Wilk tests conclude non-normality for our variables. In fact, we find at the 0.05 level (the significance level chosen in the Tanaskovic et al. (2012)), we reject the null hypothesis for every variable. For example, the chlorine ion measure (Cl.) gets a p-value of $1.566 \times 10^{-10} \ll 0.05$ so we reject the null hypothesis. For the Calcium ion variable (Ca2.), the p-value is 0.02344. Therefore we reject the null hypothesis at the 0.05 significance level but we would fail to reject the null at the 0.01 significance level had we chosen it.

And to the big question: does our data follow a multivariate normal distribution? We can run a couple of tests for this very question (significance level 0.05). The first we ran is Mardia's test which formally tests whether multiple variables considered together follows a multivariate normal distribution. There are a few things we need to have for Mardia's test to work its best. First we want independent observations, we don't want extreme outliers, and also want a reasonable sample size. In this case, we should only worry about the size of our sample, 30 observations is definitely smaller. Mardia's test has the null hypothesis the data follows a multivariate normal distribution and looks at two features of the data: skewness and kurtosis. Essentially whether the overall distribution is asymmetric, and whether the overall distribution has heavy or light tails.



```
## Call: mardia(x = hydroclean)
##
## Mardia tests of multivariate skew and kurtosis
## Use describe(x) the to get univariate tests
## n.obs = 30   num.vars = 12
## b1p = 133.38   skew = 666.9   with probability <= 4.3e-20
## small sample skew = 744.68   with probability <= 2.3e-28
## b2p = 199.48   kurtosis = 4.7   with probability <= 2.6e-06
```

The p-value for skewness is 4.3×10^{-20} and the p-value for kurtosis is 2.6×10^{-6} . Both of these p-values are

less than 0.05, the significance level, and so we reject the null hypothesis and say there is evidence against the data following a multivariate normal distribution.

<https://metricgate.com/docs/mardias-test/>

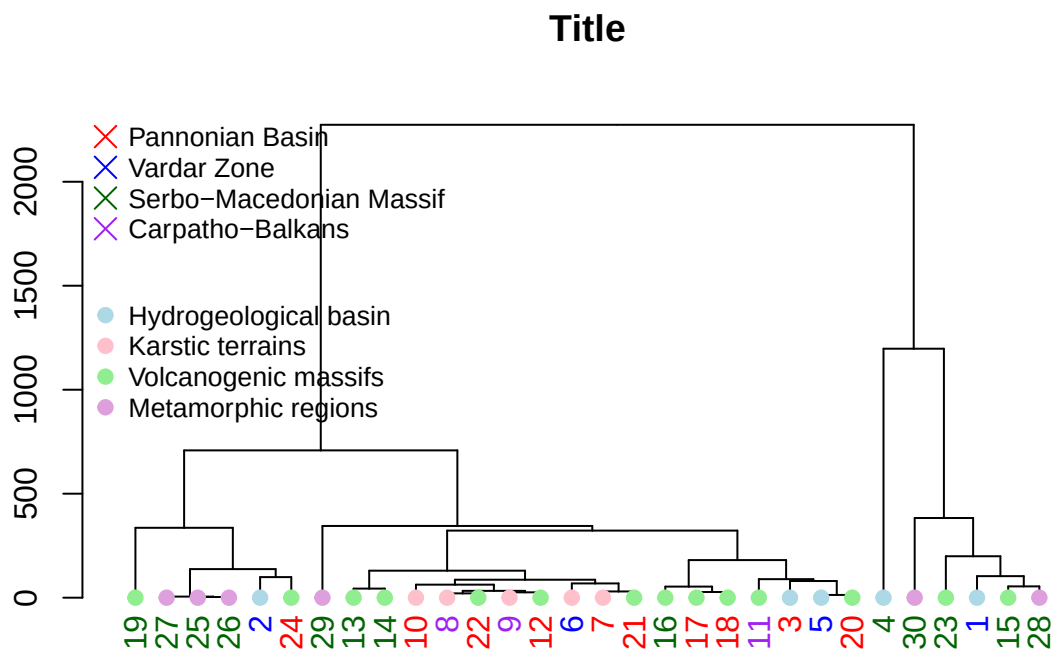
These p-values are very small but for extra confidence, especially given our small sample size, we can also do a Henze-Zirkler test with null hypothesis the data follows a multivariate normal distribution. We want independent observations, numeric variables, and it is designed for multiple variables jointly. The Henze-Zirkler test can be for smaller data sets but 30 observations is still small.

```
##           Henze-Zirkler test for Multivariate Normality
##
##  data : hydroclean
##
##  HZ           : 1.305952
##  p-value      : 0
##
##  Result   : Data are not multivariate normal (sig.level = 0.05)
```

We get a p-value of 0 from our Henze-Zirkler test and so reject the null hypothesis. So both Mardia's and the Henze-Zirkler test have evidence against multivariate normality. It would be safe to say that our data does not follow a multivariate normal distribution.

<https://metricgate.com/docs/henze-zirkler-test/>

Dendrogram



Cleaning data

Remove the first row and the first 2 columns (number, place name).

Consider data as numeric, round to 3 decimals except the class columns.

Show the data.

```
##      T  pH  EC   TS  Ca2. Mg2.   Na.   K.   Cl. SO4_2. HCO3_. SiO2
## 2 45.0 7.7 3770 4.173   5.6  3.8 1157.0  7.9  56.8      2 2903.6 36.7
## 3 25.0 7.3 5100 6.200  24.0 24.0  476.0  4.0 369.0      2  829.0 25.0
## 4 25.0 6.6  677 0.641 120.0 36.6  200.0 22.2  89.0     55  890.0 18.0
## 5 48.0 7.0 2219 2.119  60.0 18.0 2280.0 30.0 1560.0     2 3360.0 20.0
## 6 14.4 7.5 1100 1.100  56.0 48.8  226.1  3.5 198.8     82  573.4 22.0
## 7 33.0 7.4  722 0.538  60.0 67.0   69.0  7.1  28.0     90  555.0 10.0
##      Geological.structure                Region
## 2 Hydrogeological basin                Pannonian Basin
## 3 Hydrogeological basin                Pannonian Basin
## 4 Hydrogeological basin                Vardar zone
## 5 Hydrogeological basin Serbo-Macedonian massif
## 6 Hydrogeological basin Serbo-Macedonian massif
## 7      Karstic terrains                Pannonian Basin
```

```
##      T                pH                EC                TS
## Min.   :14.40   Min.   :6.500   Min.   : 110.0   Min.   :0.1000
## 1st Qu.:25.12   1st Qu.:6.825   1st Qu.: 419.8   1st Qu.:0.3977
## Median :30.50   Median :7.250   Median :1100.0   Median :1.0450
## Mean   :34.07   Mean   :7.303   Mean   :1433.7   Mean   :1.4420
## 3rd Qu.:39.77   3rd Qu.:7.500   3rd Qu.:2174.2   3rd Qu.:2.0642
## Max.   :95.00   Max.   :9.000   Max.   :5100.0   Max.   :6.2000
##      Ca2.                Mg2.                Na.                K.
## Min.   :  4.00   Min.   :  0.10   Min.   :   6.40   Min.   :  0.20
## 1st Qu.: 32.00   1st Qu.: 18.07   1st Qu.:  53.55   1st Qu.:  3.25
## Median : 50.00   Median : 27.75   Median : 253.20   Median :  9.15
## Mean   : 61.02   Mean   : 44.46   Mean   : 447.09   Mean   :16.58
## 3rd Qu.: 78.50   3rd Qu.: 60.25   3rd Qu.: 468.75   3rd Qu.:30.75
## Max.   :178.00   Max.   :293.00   Max.   :2280.00   Max.   :47.00
##      Cl.                SO4_2.                HCO3_.                SiO2
## Min.   :   7.00   Min.   :  0.90   Min.   :   79.3   Min.   :   2.00
## 1st Qu.:  18.00   1st Qu.:  5.25   1st Qu.: 384.5   1st Qu.: 20.00
## Median :  33.25   Median : 17.00   Median : 859.5   Median : 35.00
## Mean   : 108.70   Mean   : 57.53   Mean   :1296.9   Mean   : 38.73
## 3rd Qu.:  73.00   3rd Qu.: 54.50   3rd Qu.:2111.2   3rd Qu.: 45.00
## Max.   :1560.00   Max.   :486.00   Max.   :3538.0   Max.   :120.00
##      Geological.structure                Region
## Hydrogeological basin: 5                Carphato-Balcans                : 3
## Karstic terrains      : 5                Pannonian Basin                : 3
## Metamorphic regions   : 6                Serbo-Macedonian massif:14
## Volcanogenic massifs  :14                Vardar zone                    :10
##
##
```

Remove the last 2 columns to get the raw numerical data.

```
##      T  pH  EC   TS  Ca2. Mg2.   Na.   K.   Cl. SO4_2. HCO3_. SiO2
## 2 45.0 7.7 3770 4.173   5.6  3.8 1157.0  7.9  56.8      2 2903.6 36.7
## 3 25.0 7.3 5100 6.200  24.0 24.0  476.0  4.0 369.0      2  829.0 25.0
## 4 25.0 6.6  677 0.641 120.0 36.6  200.0 22.2  89.0     55  890.0 18.0
```

```
## 5 48.0 7.0 2219 2.119 60.0 18.0 2280.0 30.0 1560.0      2 3360.0 20.0
## 6 14.4 7.5 1100 1.100 56.0 48.8 226.1 3.5 198.8      82 573.4 22.0
## 7 33.0 7.4 722 0.538 60.0 67.0 69.0 7.1 28.0      90 555.0 10.0
```

```
##          T          pH          EC          TS
## Min.    :14.40   Min.    :6.500   Min.    : 110.0   Min.    :0.1000
## 1st Qu.:25.12   1st Qu.:6.825   1st Qu.: 419.8   1st Qu.:0.3977
## Median :30.50   Median :7.250   Median :1100.0   Median :1.0450
## Mean    :34.07   Mean    :7.303   Mean    :1433.7   Mean    :1.4420
## 3rd Qu.:39.77   3rd Qu.:7.500   3rd Qu.:2174.2   3rd Qu.:2.0642
## Max.    :95.00   Max.    :9.000   Max.    :5100.0   Max.    :6.2000
##      Ca2.      Mg2.      Na.      K.
## Min.    : 4.00   Min.    : 0.10   Min.    : 6.40   Min.    : 0.20
## 1st Qu.: 32.00   1st Qu.: 18.07   1st Qu.: 53.55   1st Qu.: 3.25
## Median : 50.00   Median : 27.75   Median : 253.20   Median : 9.15
## Mean    : 61.02   Mean    : 44.46   Mean    : 447.09   Mean    :16.58
## 3rd Qu.: 78.50   3rd Qu.: 60.25   3rd Qu.: 468.75   3rd Qu.:30.75
## Max.    :178.00   Max.    :293.00   Max.    :2280.00   Max.    :47.00
##      Cl.      S04_2.      HC03_      Si02
## Min.    : 7.00   Min.    : 0.90   Min.    : 79.3   Min.    : 2.00
## 1st Qu.: 18.00   1st Qu.: 5.25   1st Qu.: 384.5   1st Qu.: 20.00
## Median : 33.25   Median : 17.00   Median : 859.5   Median : 35.00
## Mean    : 108.70   Mean    : 57.53   Mean    :1296.9   Mean    : 38.73
## 3rd Qu.: 73.00   3rd Qu.: 54.50   3rd Qu.:2111.2   3rd Qu.: 45.00
## Max.    :1560.00   Max.    :486.00   Max.    :3538.0   Max.    :120.00
```

Correlation Analysis

```
##          T          pH          EC          TS          Ca2.
## T      1.00000000 -0.048279355  0.01392673 -0.018543762 -0.362587398
## pH     -0.04827935  1.000000000 -0.30117234 -0.276944829 -0.446971994
## EC      0.01392673 -0.301172342  1.000000000  0.959800431  0.128555643
## TS     -0.01854376 -0.276944829  0.959800431  1.000000000  0.006757318
## Ca2.   -0.36258740 -0.446971994  0.12855564  0.006757318  1.000000000
## Mg2.   -0.34884931 -0.475968401  0.15332055  0.127391631  0.415752328
## Na.     0.13893669 -0.197114627  0.59170271  0.487667641  0.109619158
## K.     -0.05497932 -0.460240746  0.44367411  0.374611587  0.497683099
## Cl.     0.13009795 -0.112109864  0.27660626  0.250051971 -0.001765494
## S04_2.  0.31871169 -0.070251719  0.23949387 -0.006203668  0.294521256
## HC03_   -0.04471340 -0.359897943  0.62293359  0.530247294  0.320487529
## Si02    0.37018940 -0.006208081  0.15577500  0.154258218 -0.093496027
##      Mg2.      Na.      K.      Cl.      S04_2.      HC03_
## T     -0.34884931  0.1389367 -0.05497932  0.130097953  0.318711692 -0.0447134
## pH    -0.47596840 -0.1971146 -0.46024075 -0.112109864 -0.070251719 -0.3598979
## EC     0.15332055  0.5917027  0.44367411  0.276606256  0.239493868  0.6229336
## TS     0.12739163  0.4876676  0.37461159  0.250051971 -0.006203668  0.5302473
## Ca2.   0.41575233  0.1096192  0.49768310 -0.001765494  0.294521256  0.3204875
## Mg2.   1.00000000 -0.1000779  0.18818539 -0.096741715 -0.054072668  0.2304716
## Na.   -0.10007793  1.0000000  0.65093159  0.665402853  0.209176940  0.8772456
## K.     0.18818539  0.6509316  1.00000000  0.141665266  0.154814164  0.8268152
## Cl.   -0.09674171  0.6654029  0.14166527  1.000000000 -0.037284119  0.3500227
## S04_2. -0.05407267  0.2091769  0.15481416 -0.037284119  1.000000000  0.1304865
## HC03_  0.23047165  0.8772456  0.82681516  0.350022709  0.130486531  1.0000000
```

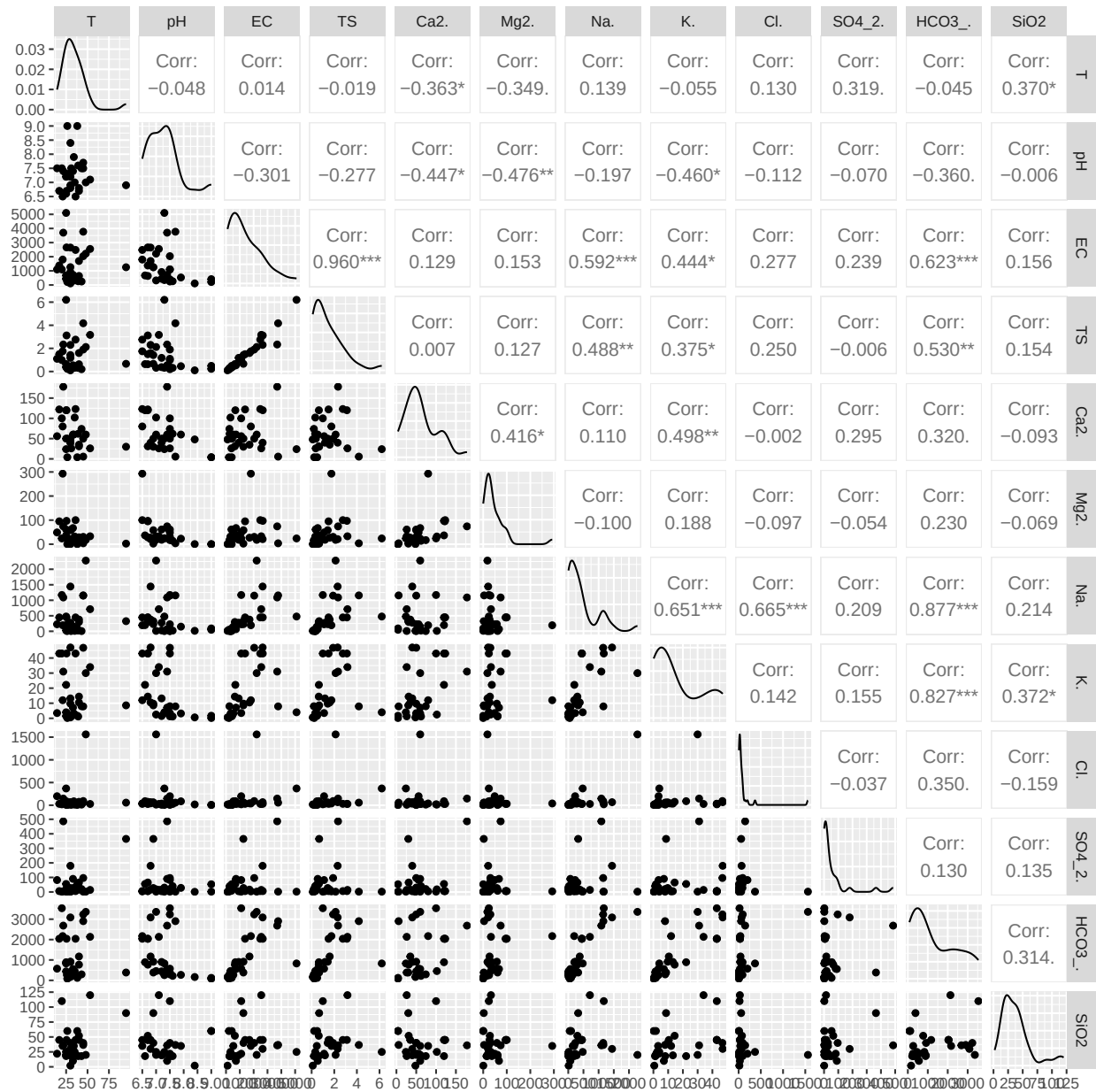


```

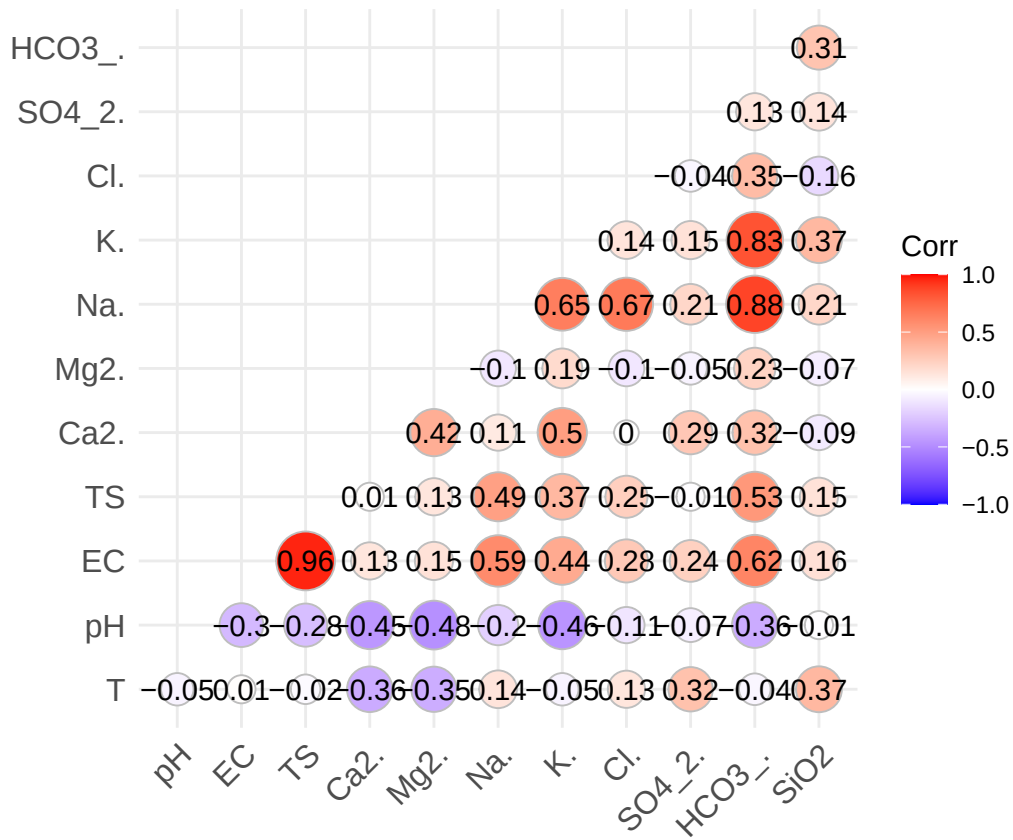
## SiO2    -0.06911767  0.2141303  0.37229615 -0.159129253  0.135391338  0.3137873
##          SiO2
## T        0.370189405
## pH       -0.006208081
## EC        0.155774998
## TS        0.154258218
## Ca2.     -0.093496027
## Mg2.     -0.069117668
## Na.       0.214130329
## K.        0.372296154
## Cl.      -0.159129253
## SO4_2.   0.135391338
## HCO3_    0.313787294
## SiO2     1.000000000

```

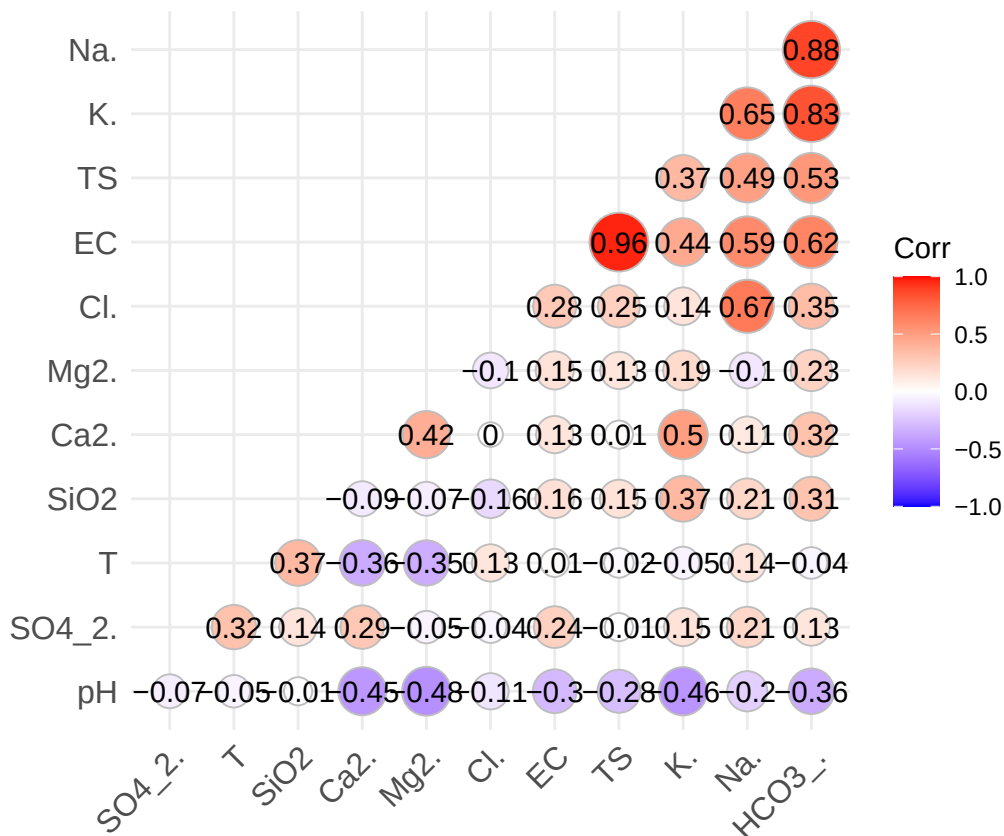
Pairs plot



Correlogram



Correlogram, hierarchically ordered



TODO: analysing correlations between compounds

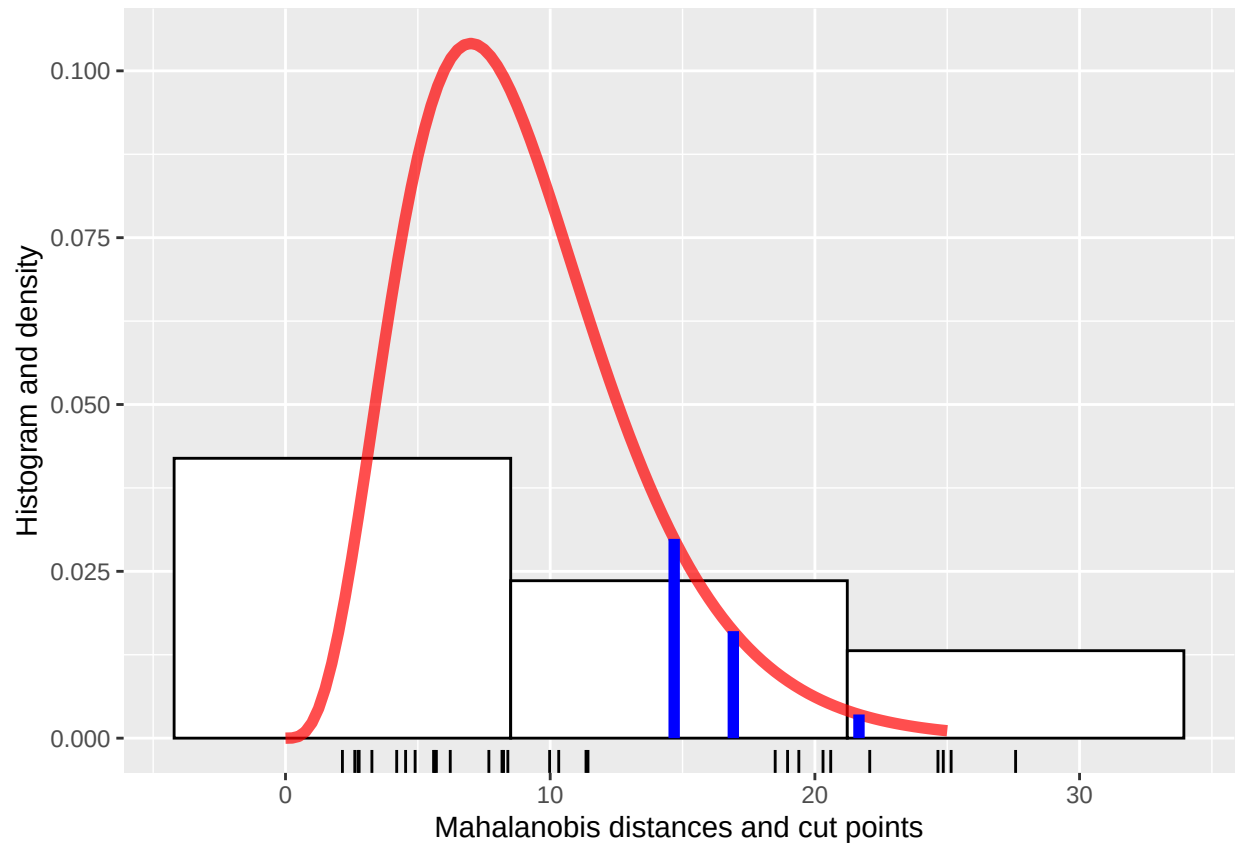
Multivariate Outlier Assessment (Mahalanobis Distance)

Mahalanobis distance for assessing measurements multivariately.

##	T	pH	EC	TS	Ca2.	Mg2.
##	34.073333	7.303333	1433.733333	1.441967	61.020000	44.456667
##	Na.	K.	Cl.	SO4_2.	HCO3_.	SiO2
##	447.086667	16.583333	108.703333	57.530000	1296.913333	38.733333

##	T	pH	EC	TS	Ca2.
## T	221.3351264	-0.4568046	258.5789	-0.3828078	-222.7187586
## pH	-0.4568046	0.4044713	-239.0439	-0.2443964	-11.7366207
## EC	258.5788506	-239.0439080	1557533.9264	1662.1012322	6624.1296552
## TS	-0.3828078	-0.2443964	1662.1012	1.9253760	0.3871248
## Ca2.	-222.7187586	-11.7366207	6624.1297	0.3871248	1704.6602759
## Mg2.	-285.8601609	-16.6729540	10539.2501	9.7361951	945.4622759
## Na.	1117.3351494	-67.7647816	399175.3101	365.7826202	2446.5085517
## K.	-13.6704598	-4.8920115	9254.2540	8.6875546	343.4237931
## Cl.	548.8576782	-20.2186322	97891.4733	98.3902725	-20.6704138
## SO4_2.	515.1253103	-4.8538966	32471.5393	-0.9351817	1321.0683448

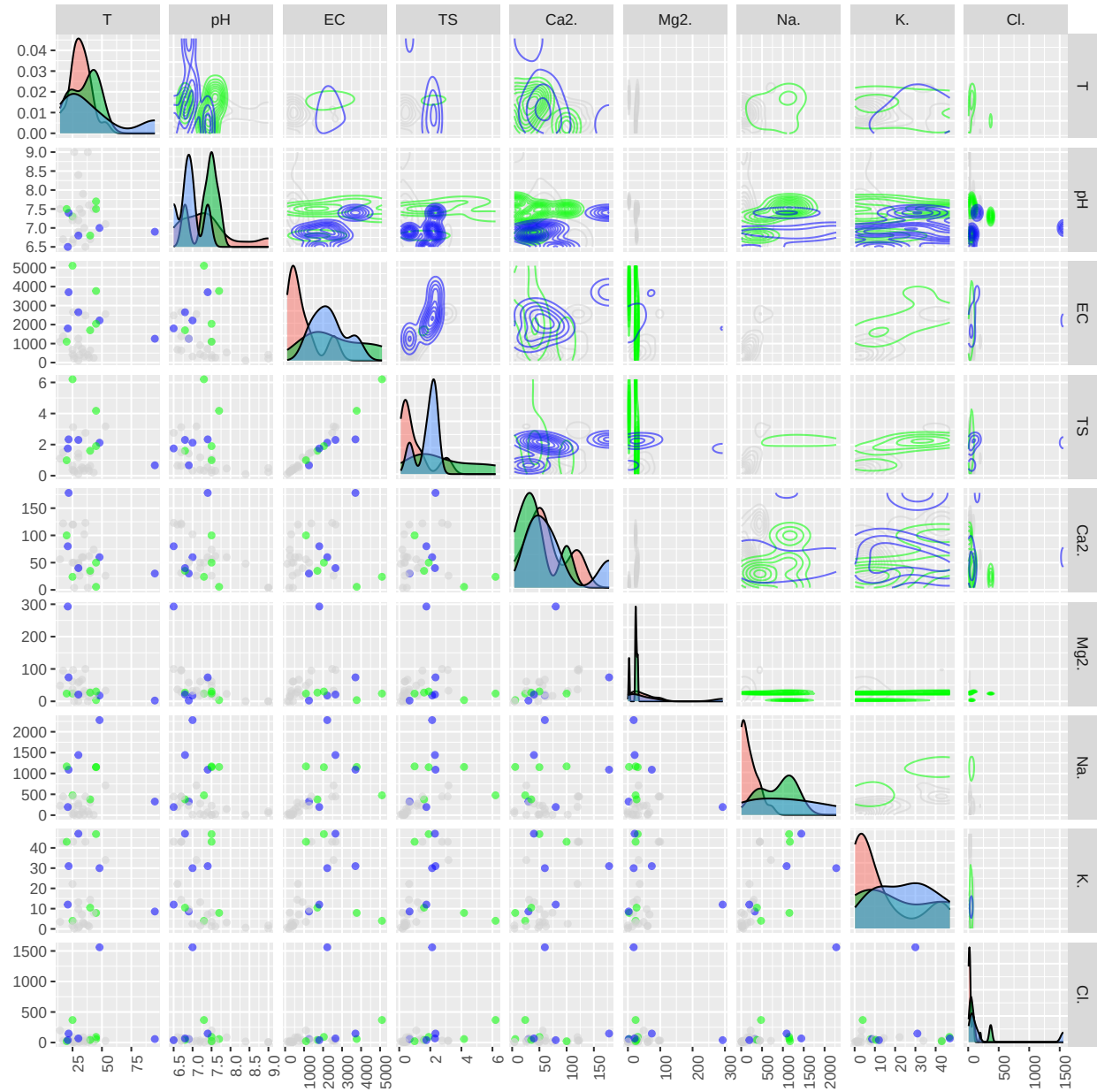
##	HC03_.	-756.0837701	-260.1538391	883624.0830	836.2637522	15039.6280000
##	Si02	150.2340230	-0.1077011	5303.1713	5.8388218	-105.3006897
##		Mg2.	Na.	K.	Cl.	S04_2.
##	T	-285.860161	1117.33515	-13.670460	548.85768	515.1253103
##	pH	-16.672954	-67.76478	-4.892011	-20.21863	-4.8538966
##	EC	10539.250115	399175.31011	9254.254023	97891.47333	32471.5393103
##	TS	9.736195	365.78262	8.687555	98.39027	-0.9351817
##	Ca2.	945.462276	2446.50855	343.423793	-20.67041	1321.0683448
##	Mg2.	3033.759782	-2979.68439	173.234770	-1511.01399	-323.5624483
##	Na.	-2979.684391	292201.67982	5880.784598	101997.63970	12284.1435172
##	K.	173.234770	5880.78460	279.329713	671.40730	281.0987931
##	Cl.	-1511.013989	101997.63970	671.407299	80413.36171	-1148.6235517
##	S04_2.	-323.562448	12284.14352	281.098793	-1148.62355	11802.6476897
##	HC03_.	14428.299218	538976.20018	15706.305402	112815.13547	16112.4820000
##	Si02	-103.848161	3157.46598	169.732989	-1230.92977	401.2358621
##		HC03_.	Si02			
##	T	-756.0838	150.2340230			
##	pH	-260.1538	-0.1077011			
##	EC	883624.0830	5303.1712644			
##	TS	836.2638	5.8388218			
##	Ca2.	15039.6280	-105.3006897			
##	Mg2.	14428.2992	-103.8481609			
##	Na.	538976.2002	3157.4659770			
##	K.	15706.3054	169.7329885			
##	Cl.	112815.1355	-1230.9297701			
##	S04_2.	16112.4820	401.2358621			
##	HC03_.	1291855.9143	9728.8505747			
##	Si02	9728.8506	744.1119540			



Assessing measurements

```
##
##   Typical   Somewhat Surprising   Very
##        20         0         5         5
```

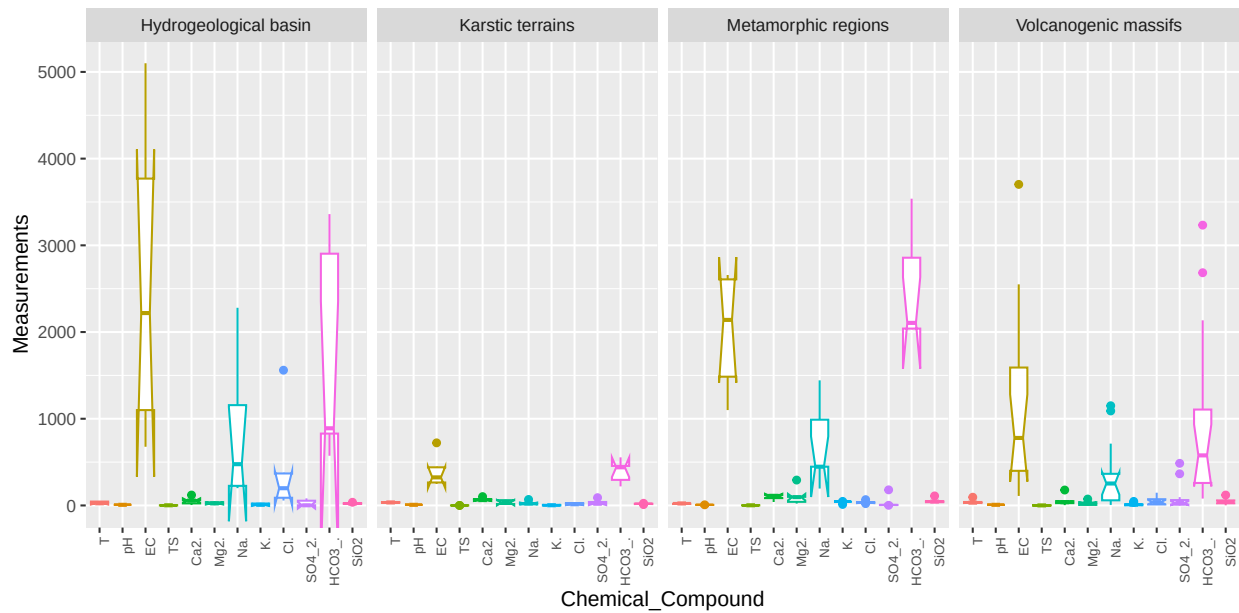
Pairs plot colored by surprise level



Analyzing the Data with Classes

Melting the data

Box plots with notches

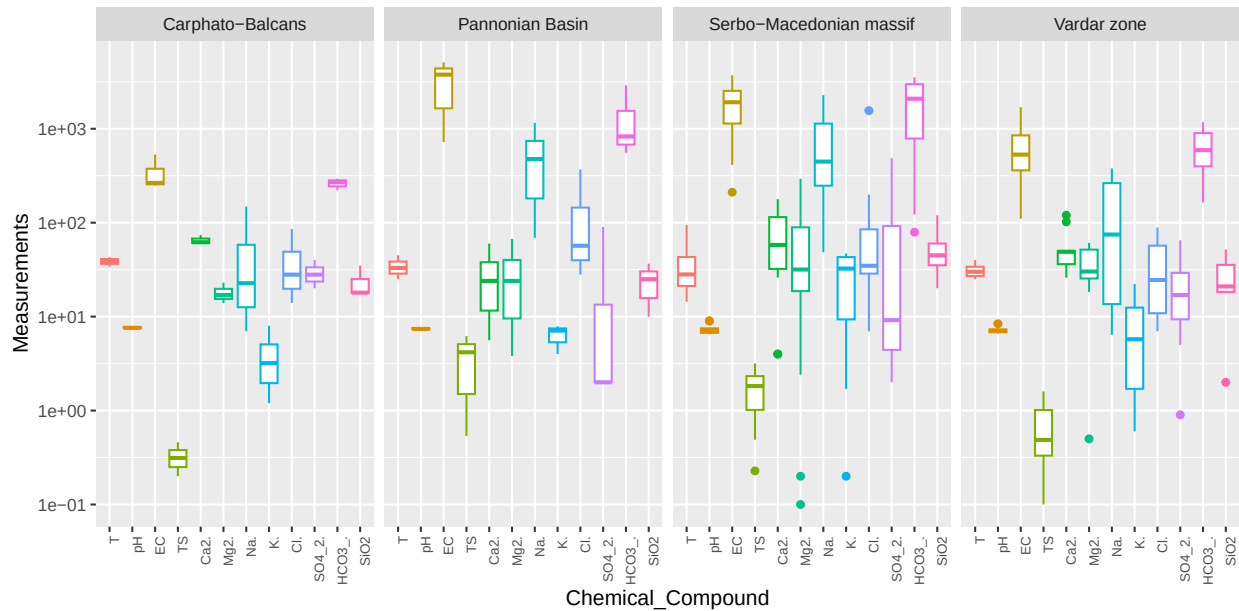


Notice, the data doesn't have enough samples for the notches to appear nicely.

Box plots, log scale geology



Box plots, log scale region



TODO: Analyse box plots

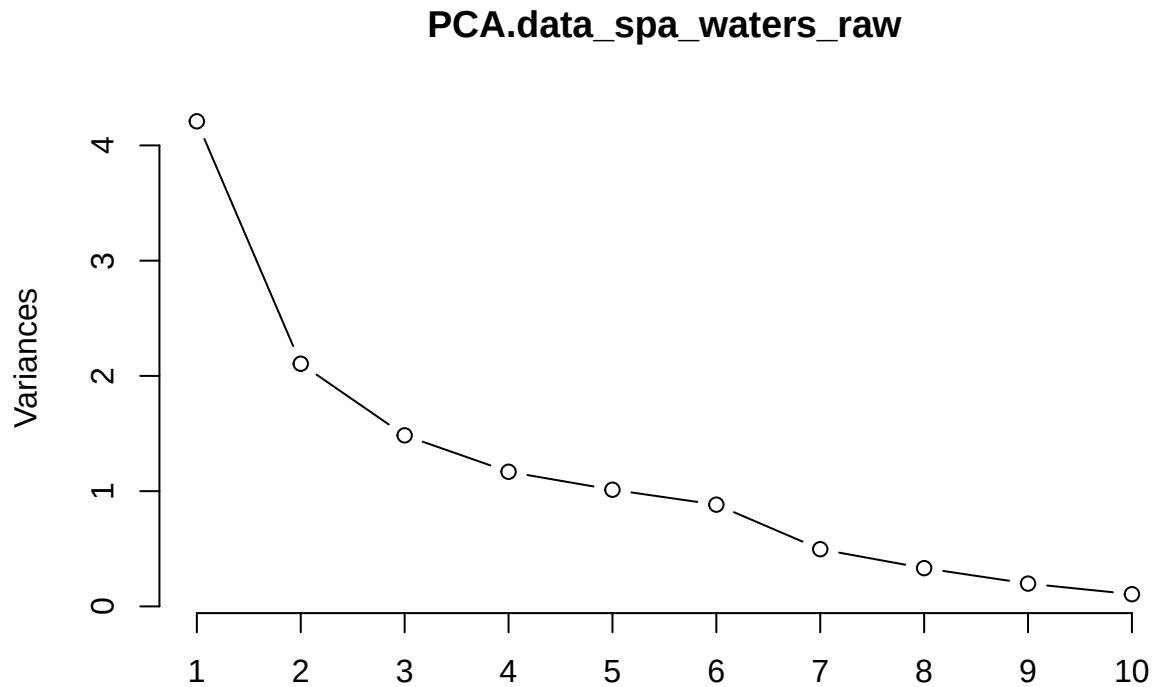
Principal Component Analysis (PCA)

Reset `data_spa_waters_raw` back to the original (without the Mahalanobis columns added earlier).

Calculate PCA

```
## Importance of components:
##          PC1    PC2    PC3    PC4    PC5    PC6    PC7
## Standard deviation  2.0516 1.4511 1.2182 1.08088 1.00623 0.93965 0.70463
## Proportion of Variance 0.3508 0.1755 0.1237 0.09736 0.08437 0.07358 0.04138
## Cumulative Proportion 0.3508 0.5262 0.6499 0.74727 0.83164 0.90522 0.94660
##          PC8    PC9    PC10    PC11    PC12
## Standard deviation  0.57619 0.4450 0.32543 0.05581 0.04290
## Proportion of Variance 0.02767 0.0165 0.00883 0.00026 0.00015
## Cumulative Proportion 0.97426 0.9908 0.99959 0.99985 1.00000
```

Variance explained (scree plot)



TODO: Say some words about it

PCA components

```
##      T      pH      EC      TS      Ca2.      Mg2.      Na.      K.
##  34.073   7.303 1433.733   1.442   61.020   44.457  447.087  16.583
##      Cl.   SO4_2.   HCO3_   SiO2
## 108.703   57.530 1296.913   38.733
```

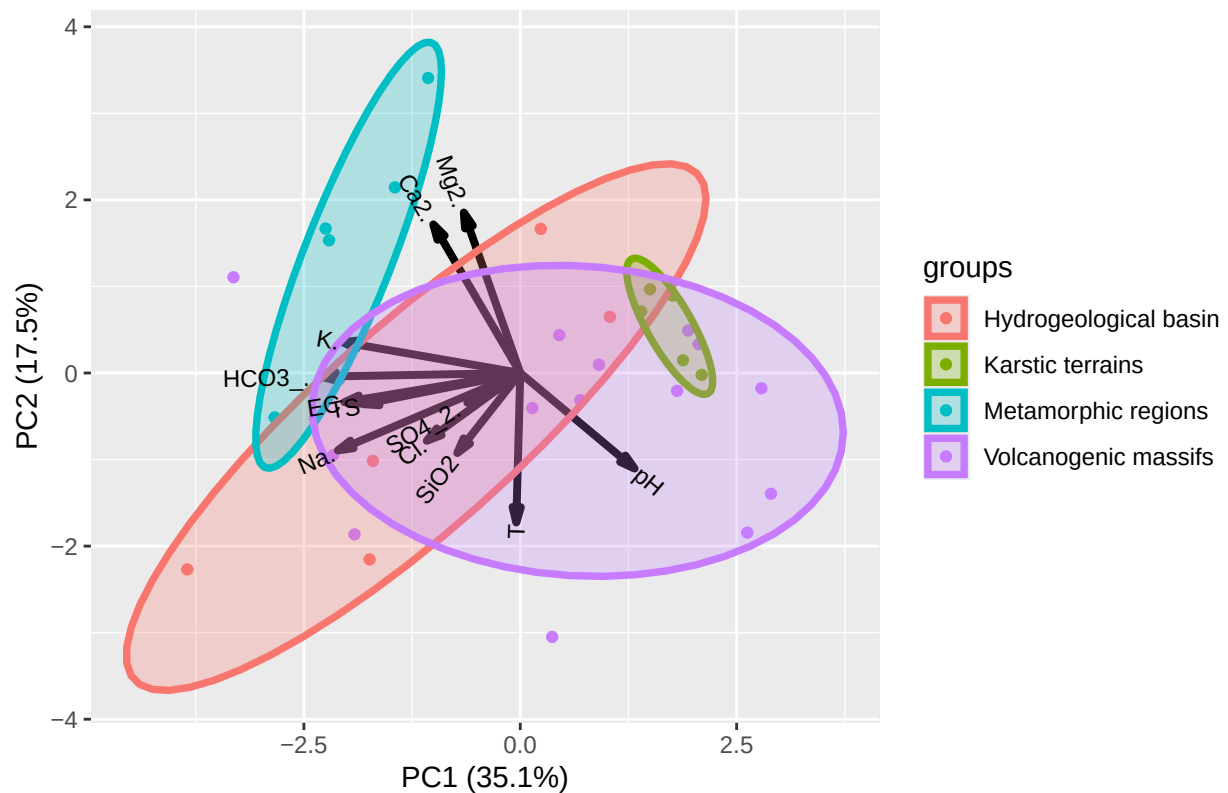
```
## [1] 2.05162 1.45109 1.21823 1.08088 1.00623 0.93965 0.70463 0.57619 0.44497
## [10] 0.32543 0.05581 0.04290
```

```
##      PC1      PC2      PC3      PC4      PC5      PC6      PC7
## T      -0.008934 -0.47274 -0.359101 -0.05022 -0.25540  0.47386 -0.03026
## pH      0.255693 -0.30222  0.138232  0.01108  0.22154 -0.60931  0.39727
## EC     -0.393539 -0.09217  0.167864  0.30805 -0.36906 -0.20141 -0.02824
## TS     -0.350421 -0.09868  0.277014  0.44621 -0.28642 -0.09763 -0.14653
## Ca2.    -0.194533  0.46994 -0.228328 -0.31475 -0.02988 -0.20700 -0.18009
## Mg2.    -0.126284  0.50744 -0.014274  0.19263 -0.10320  0.27981  0.71540
## Na.     -0.408190 -0.24496  0.118662 -0.27672  0.16960 -0.04761  0.10770
## K.      -0.398679  0.09808 -0.212842 -0.04847  0.36067 -0.04693 -0.27182
## Cl.     -0.209239 -0.21442  0.406407 -0.50456 -0.02990  0.27464  0.25275
```

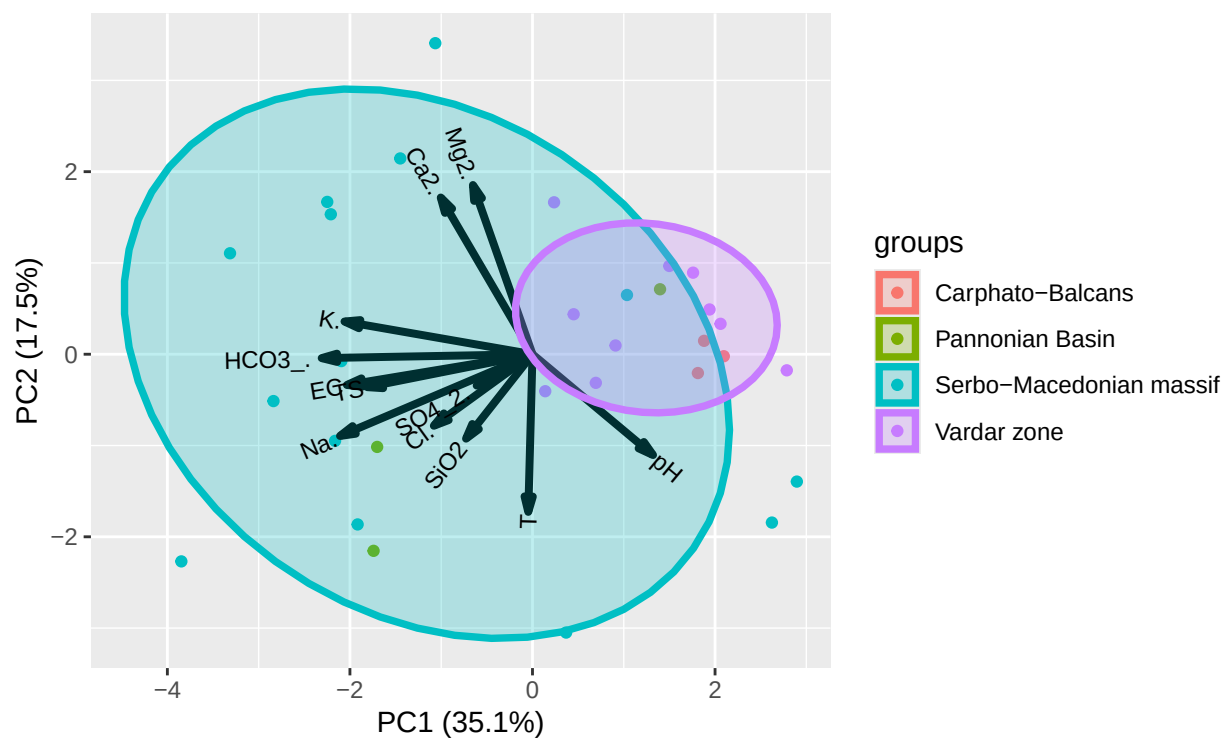
##	SO4_2.	-0.118433	-0.09385	-0.495788	-0.29202	-0.52376	-0.38070	0.22767
##	HC03_.	-0.447053	-0.01157	-0.003406	-0.04232	0.28031	-0.05738	0.13516
##	SiO2	-0.140906	-0.25312	-0.473624	0.38289	0.37850	0.06952	0.23515
##		PC8	PC9	PC10	PC11	PC12		
##	T	0.06496	-0.58594	-0.07642	0.029763	0.0002768		
##	pH	-0.06943	-0.47765	0.11292	0.006804	0.0107307		
##	EC	-0.07673	-0.04082	-0.02709	0.274679	-0.6725015		
##	TS	-0.17414	-0.11074	0.07775	-0.235970	0.6117771		
##	Ca2.	-0.52331	-0.36821	-0.30633	0.082368	0.0622618		
##	Mg2.	0.11761	-0.17904	0.13761	0.132739	0.0779402		
##	Na.	0.21027	0.12376	-0.23563	0.643630	0.3328410		
##	K.	0.16503	-0.19327	0.71382	0.022639	-0.0545398		
##	Cl.	-0.45265	0.09649	0.25837	-0.236596	-0.1155556		
##	SO4_2.	0.13071	0.29897	0.16971	-0.167292	0.1083103		
##	HC03_.	0.35292	-0.08652	-0.44426	-0.586764	-0.1561418		
##	SiO2	-0.49908	0.29684	-0.06226	0.022453	-0.0134888		

Biplots

Biplot grouped by Geological structure.

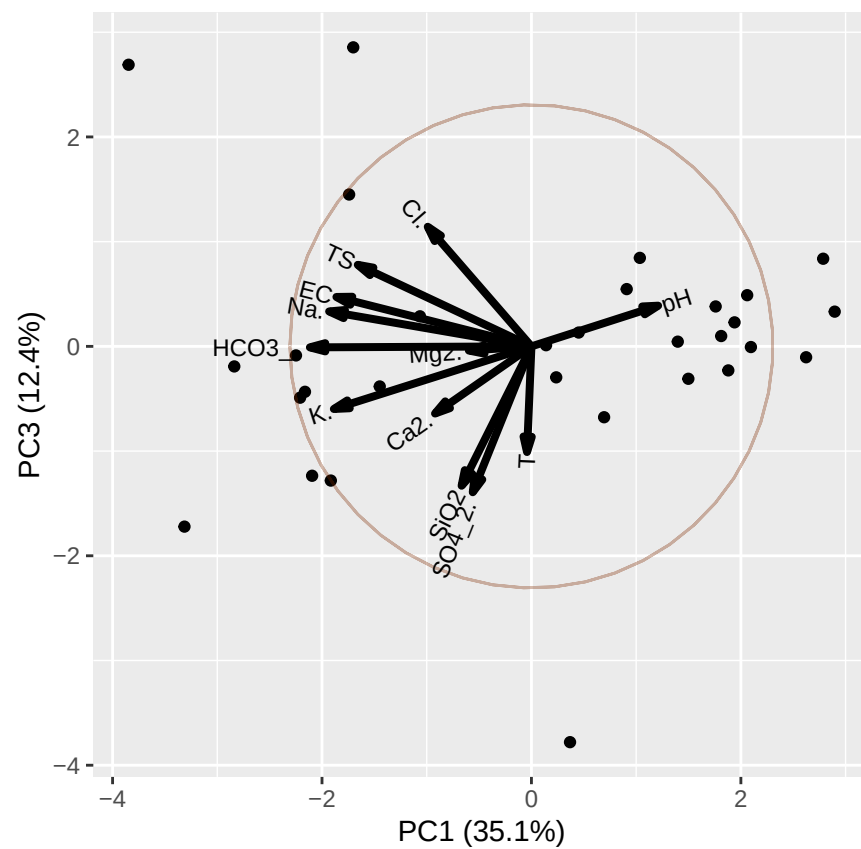


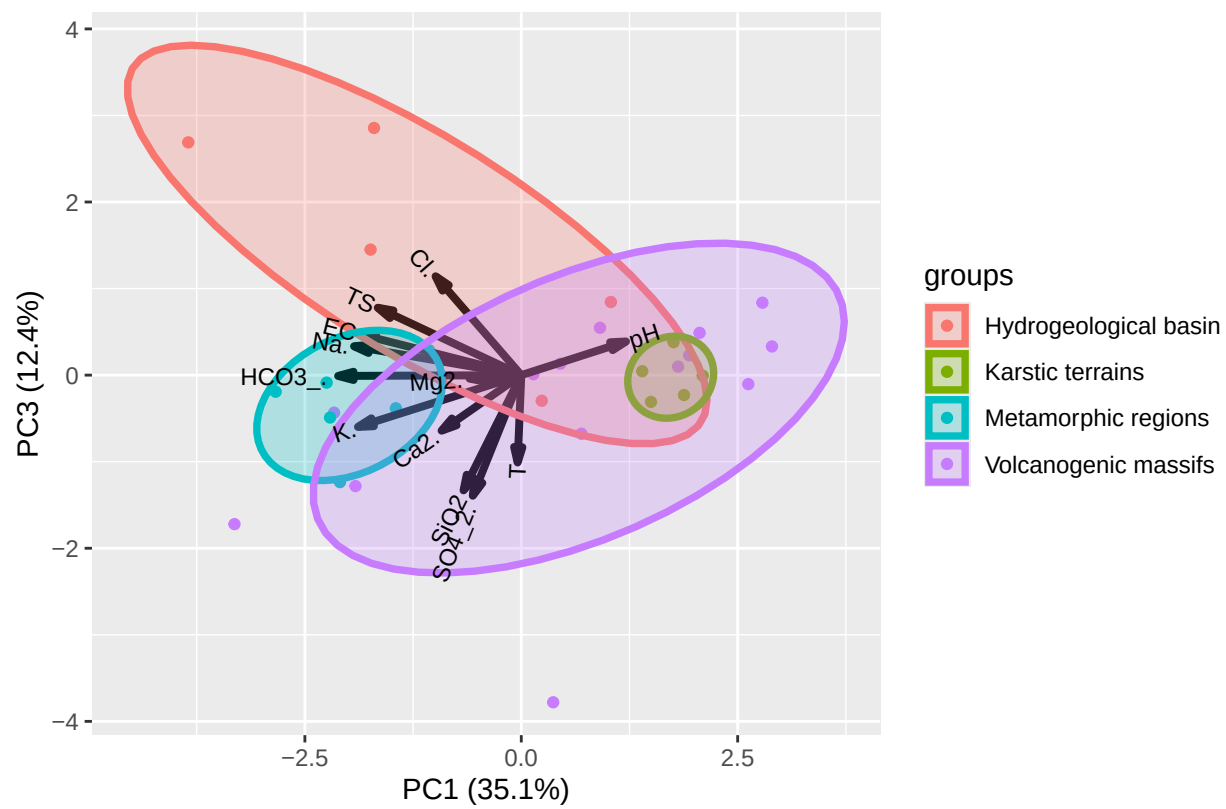
Biplot grouped by Region.

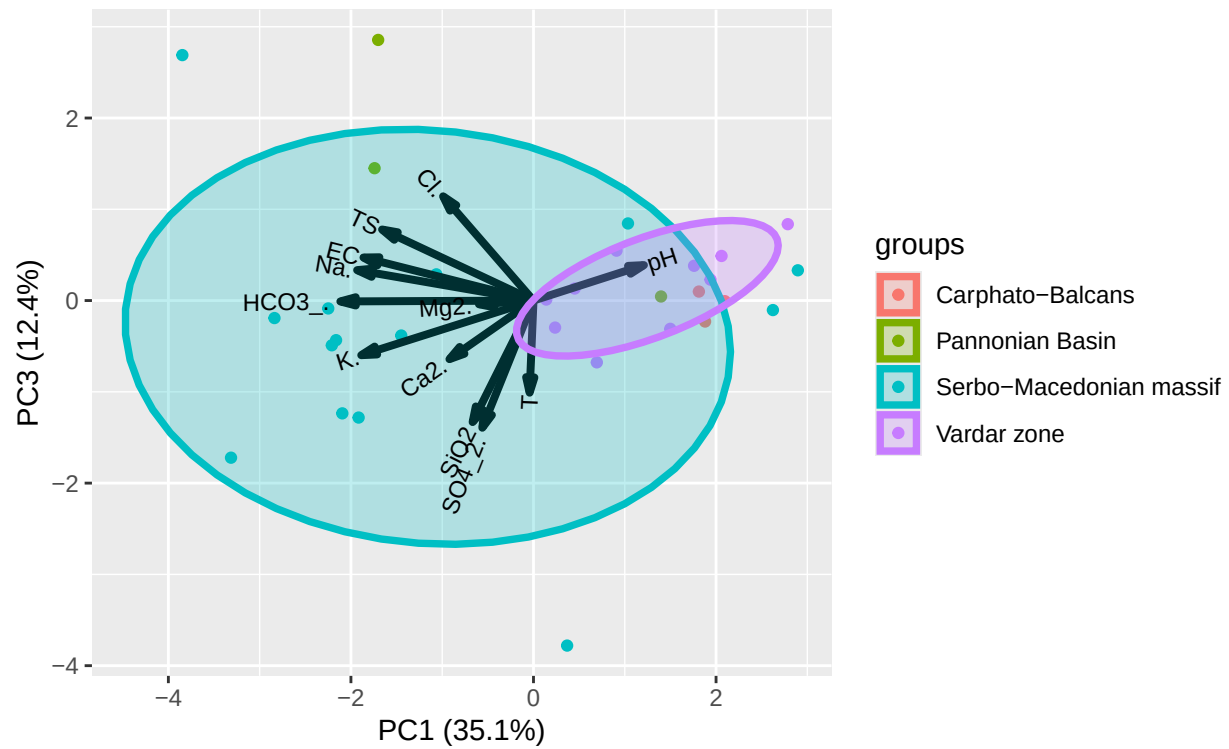


##	PC1	PC2	PC3	PC4
##	Min. : -3.913	Min. : -3.101	Min. : -3.844	Min. : -3.476
##	1st Qu.: -1.904	1st Qu.: -0.854	1st Qu.: -0.371	1st Qu.: -0.450
##	Median : 0.417	Median : 0.038	Median : 0.002	Median : -0.060
##	Mean : 0.000	Mean : 0.000	Mean : 0.000	Mean : 0.000
##	3rd Qu.: 1.830	3rd Qu.: 0.863	3rd Qu.: 0.375	3rd Qu.: 0.551
##	Max. : 2.946	Max. : 3.466	Max. : 2.904	Max. : 2.209
##	PC5	PC6	PC7	PC8
##	Min. : -2.1748	Min. : -2.9907	Min. : -1.069	Min. : -0.926
##	1st Qu.: -0.2617	1st Qu.: -0.6062	1st Qu.: -0.512	1st Qu.: -0.397
##	Median : -0.0473	Median : 0.0464	Median : -0.118	Median : -0.116
##	Mean : 0.0000	Mean : 0.0000	Mean : 0.000	Mean : 0.000
##	3rd Qu.: 0.2152	3rd Qu.: 0.5104	3rd Qu.: 0.290	3rd Qu.: 0.245
##	Max. : 3.1276	Max. : 1.6797	Max. : 2.551	Max. : 1.688
##	PC9	PC10	PC11	PC12
##	Min. : -0.8072	Min. : -1.0115	Min. : -0.14200	Min. : -0.16870
##	1st Qu.: -0.2828	1st Qu.: -0.1203	1st Qu.: -0.02465	1st Qu.: -0.01622
##	Median : 0.0026	Median : 0.0761	Median : -0.00464	Median : 0.00669
##	Mean : 0.0000	Mean : 0.0000	Mean : 0.00000	Mean : 0.00000
##	3rd Qu.: 0.3474	3rd Qu.: 0.2250	3rd Qu.: 0.01628	3rd Qu.: 0.02236
##	Max. : 0.8152	Max. : 0.3924	Max. : 0.15220	Max. : 0.06104

Biplot for PC1 vs PC3.







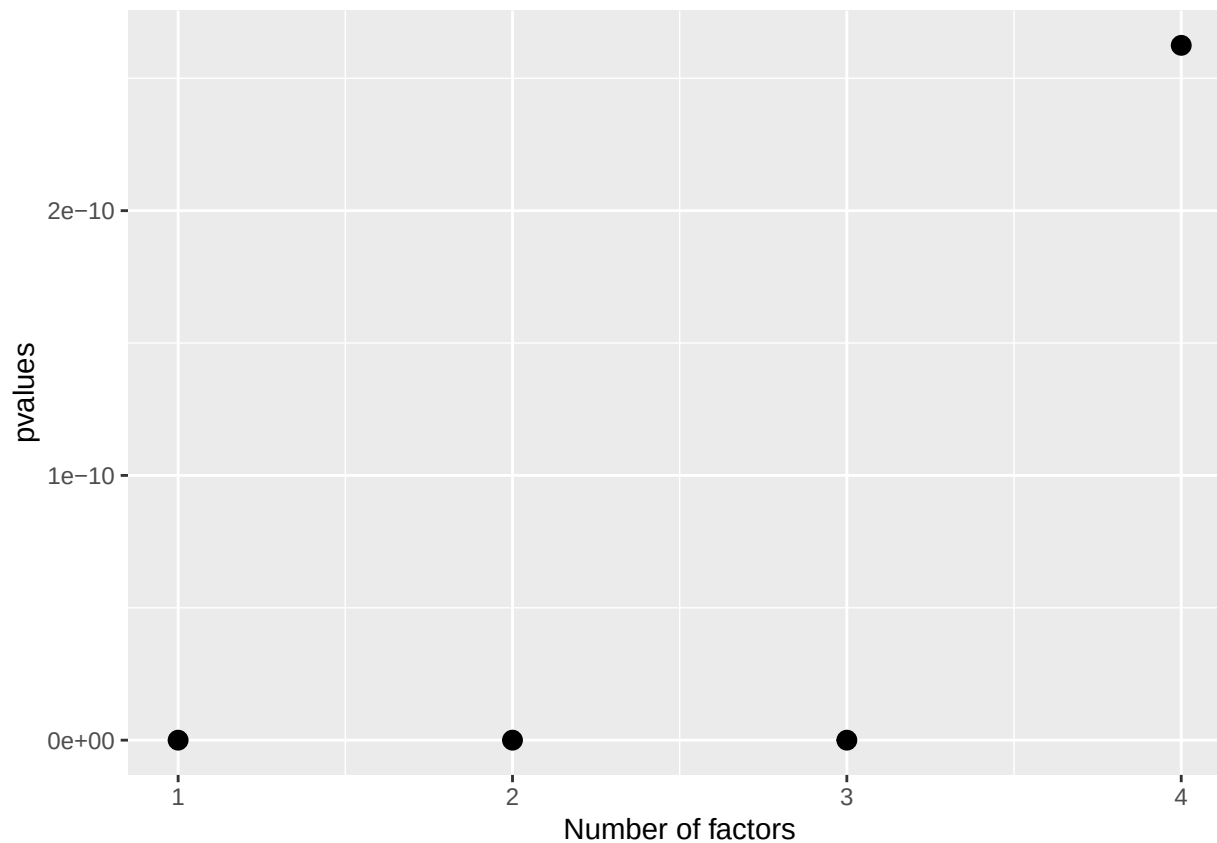
TODO: Analyse the pair plots

```
##              PC1    PC2    PC3    PC4
## Standard deviation  2.0516 1.4511 1.2182 1.08088
## Proportion of Variance 0.3508 0.1755 0.1237 0.09736
## Cumulative Proportion 0.3508 0.5262 0.6499 0.74727
```

Factor Analysis

Factor analysis with `factanal`

Choosing number of factors via p-value.

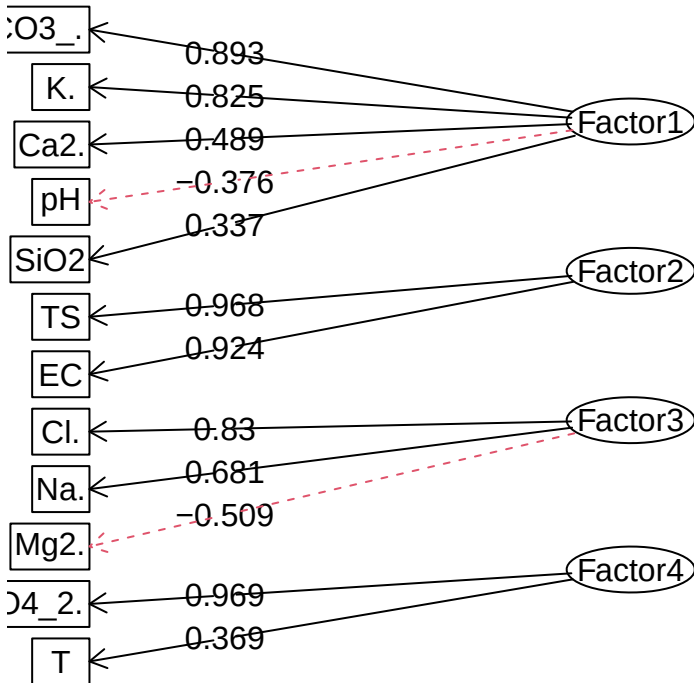


```
##
## Call:
## factanal(x = data_spa_waters_raw, factors = 4)
##
## Uniquenesses:
##      T      pH      EC      TS      Ca2.      Mg2.      Na.      K.      Cl.      S04_2.      HC03_.
## 0.789 0.783 0.005 0.005 0.632 0.560 0.005 0.274 0.280 0.005 0.005
## Si02
## 0.876
##
## Loadings:
##      Factor1 Factor2 Factor3 Factor4
## T      -0.107          0.252 0.369
## pH     -0.376 -0.220 0.162
## EC      0.330 0.924 0.114 0.140
## TS      0.199 0.968 0.103
## Ca2.    0.489          -0.315 0.170
## Mg2.    0.369 0.108 -0.509 -0.183
## Na.     0.652 0.307 0.681 0.108
## K.      0.825 0.194
## Cl.          0.164 0.830
## S04_2. 0.221          0.969
## HC03_. 0.893 0.328 0.294
## Si02    0.337
##
##      Factor1 Factor2 Factor3 Factor4
```



```
## SS loadings      2.747    2.126    1.726    1.184
## Proportion Var   0.229    0.177    0.144    0.099
## Cumulative Var   0.229    0.406    0.550    0.649
##
## Test of the hypothesis that 4 factors are sufficient.
## The chi square statistic is 94.44 on 24 degrees of freedom.
## The p-value is 2.62e-10
```

Factor Analysis



```
##      Factor1  Factor2  Factor3  Factor4
## 2  0.61609  1.79806  0.645077 -0.68643
## 3 -1.94688  3.87567  0.226146 -0.23610
## 4 -0.05781 -0.56543 -0.362431 -0.01770
## 5  0.61999 -0.08108  4.426249 -0.30038
## 6 -0.64673 -0.10122  0.002456  0.37553
## 7 -0.36272 -0.49659 -0.502985  0.37207
## 8 -0.52391 -0.63687 -0.307749 -0.27203
## 9 -0.77403 -0.69515 -0.101714  0.02645
## 10 -0.64567 -0.76668 -0.199266 -0.18296
## 11 -0.45979 -0.71664 -0.371775 -0.26379
## 12 -0.87652 -0.54208  0.258033 -0.02395
## 13 -0.71514 -0.65895 -0.205355 -0.08315
## 14 -1.01977 -0.53957  0.216086  0.22003
## 15 -1.03307 -0.71533  0.258651 -0.23364
## 16  1.82513 -0.10056  0.235872 -0.03912
## 17 -0.56013 -0.27941 -0.133954  3.01459
```

```

## 18 -0.33302 -0.56875  0.134986  0.17376
## 19 -0.35243 -0.20864  0.028951 -0.41691
## 20  0.40793  1.04040 -0.044082 -0.57688
## 21 -0.32315  0.04810 -0.263099 -0.24155
## 22 -0.59947 -0.64373 -0.266439 -0.31892
## 23 -0.85598 -0.83254  0.071139 -0.29346
## 24  1.51839  0.84072 -0.651493  3.63748
## 25 -0.21834  0.24112 -0.031455 -0.32885
## 26  0.59176  1.11948 -0.942614 -0.75253
## 27  0.99951 -0.17177 -0.751689 -0.77051
## 28  0.67449  0.87788 -0.911691 -0.74519
## 29  2.36515 -0.97339  0.299710 -1.00348
## 30  1.29278  0.14295 -1.786700 -0.94013
## 31  1.39333  0.31000  1.031134  0.90773

```

Factor Analysis based on Principal Components (Varimax rotation)

```

##          RC1    RC2    RC3    RC4
## SS loadings      2.9244 2.5013 1.7758 1.7657
## Proportion Var    0.2437 0.2084 0.1480 0.1471
## Cumulative Var    0.2437 0.4521 0.6001 0.7473
## Proportion Explained 0.3261 0.2789 0.1980 0.1969
## Cumulative Proportion 0.3261 0.6051 0.8031 1.0000

```

```

##
## Loadings:
##          RC1    RC2    RC3    RC4
## T          -0.357  0.729
## pH        -0.257 -0.655
## EC         0.875  0.151          0.182
## TS         0.930          -0.104
## Ca2.       -0.104  0.893
## Mg2.        0.164  0.650 -0.346 -0.291
## Na.         0.557  0.164  0.283  0.722
## K.          0.490  0.614  0.315  0.208
## Cl.         0.182          0.883
## S04_2.     -0.121  0.293  0.652  0.130
## HCO3_       0.661  0.454  0.212  0.394
## SiO2        0.394          0.676 -0.330

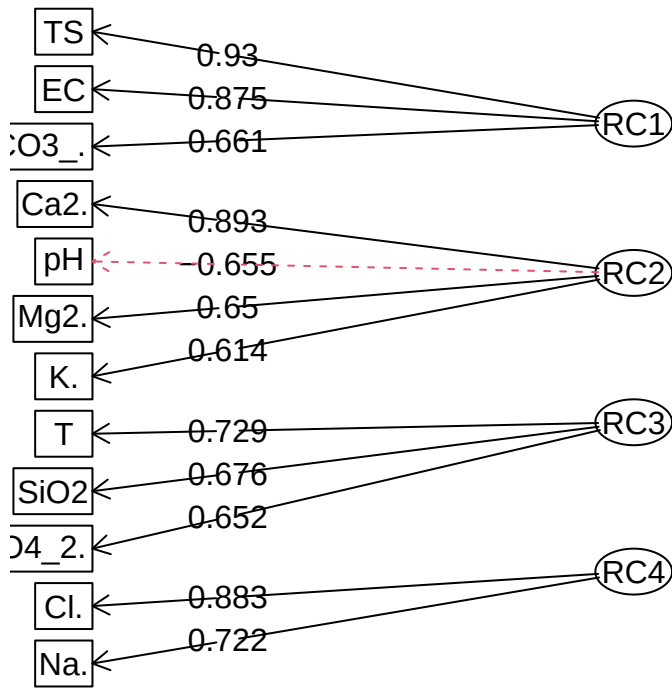
```

```

##
##          RC1    RC2    RC3    RC4
## SS loadings      2.924 2.501 1.776 1.766
## Proportion Var    0.244 0.208 0.148 0.147
## Cumulative Var    0.244 0.452 0.600 0.747

```

Factor Analysis



Score Plots

Clustering on PCA Scores

